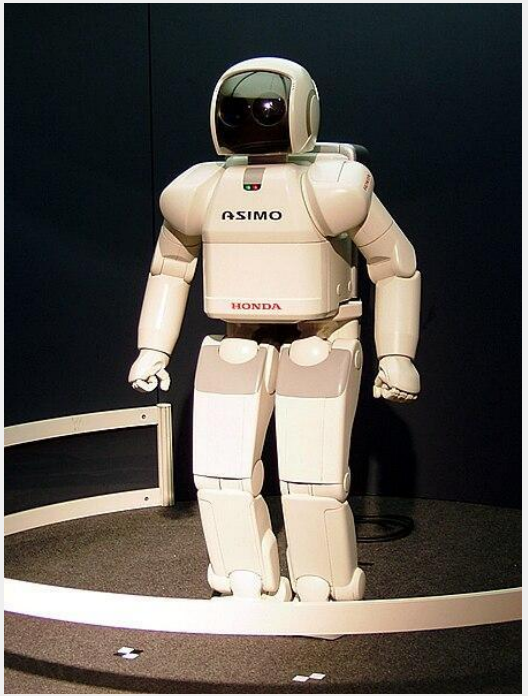
History of AI

Global Perspectives in History of AI



unsojo@cornell.edu

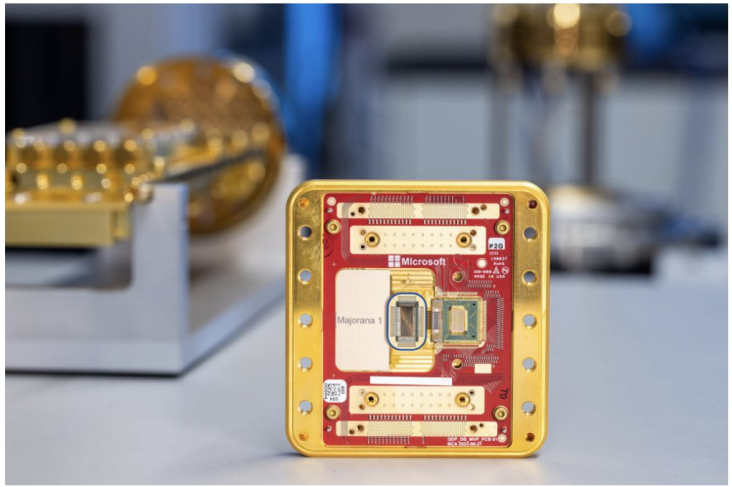
Lecture 5 Februrary 21, 2025

Trending: Microsoft quantum breakthrough promises to usher in the next era of computing in 'years, not decades'

Microsoft quantum breakthrough promises to usher in the next era of computing in 'years, not decades'

BY TODD BISHOP on February 19, 2025 at 8:00 am

Share Tweet Share Reddit Email



Microsoft's new "Majorana 1" processor (pronounced *my-or-ana*) is the first quantum chip powered by a topological core based on a new class of materials. (Photo by John Brecher for Microsoft)

Microsoft says it has developed a new quantum processor based on a novel state of matter, giving it a clear path to achieve quantum computing's long-term promise of

Microsoft believes the million-qubit goal is now years, not decades, away. Majorana 1 is the result of 20 years of research into quantum computing, the company said.

US2QC: Underexplored Systems for Utility-Scale Quantum Computing

Summary

It has been credibly hypothesized – but not proven – that quantum computers will have a transformative impact on a variety of scientific and technical disciplines.

Two separate factors make the ultimate impact of quantum computing unclear.

- First, although a number of algorithms and applications for quantum computers have been suggested, in most cases a rigorous comparison to the best classical alternatives for real-world usage has not been completed.
- Second, it is unclear when or if a "utility-scale" quantum computer – one whose computational value exceeds its costs – can be built, particularly for applications that require fault-tolerance.

The complexity of a fault-tolerant utility-scale quantum computer could approach or exceed that of

Program Manager

Joe Altepeter
Program Manager, Microsystems Techn
Office

Read bio

A Quantum Race?

BBC

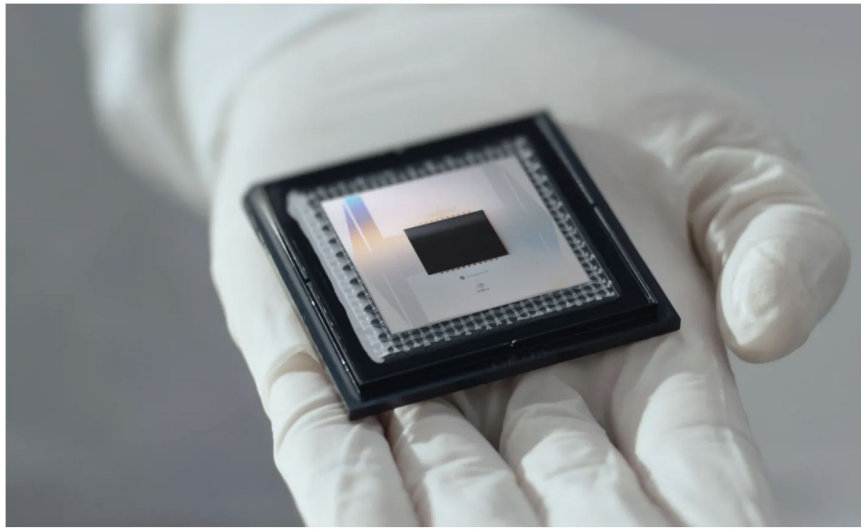
Home News Sport Business Innovation Culture Arts Travel Earth Audio Video Live

Google unveils 'mind-boggling' quantum computing chip

9 December 2024

Share Save

Chris Vallance
Senior Technology Reporter



The new Willow chip

Google has unveiled a new chip which it claims takes five minutes to solve a problem that would currently take the world's fastest super computers ten septillion – or 10,000,000,000,000,000,000,000,000 years – to complete.

The chip is the latest development in a field known as quantum computing - which is attempting to use the principles of particle physics to create a new type of mind-

IBM

IBM Newsroom News Media resources Inside IBM Blog

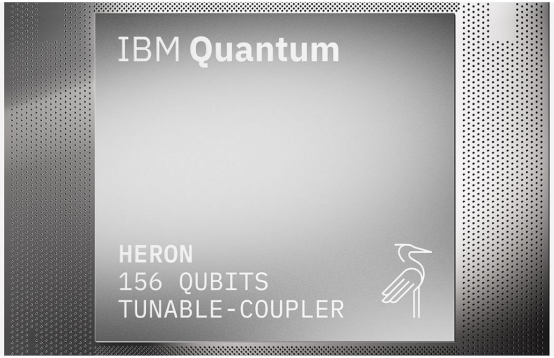
IBM Launches Its Most Advanced Quantum Computers, Fueling New Scientific Value and Progress towards Quantum Advantage

Qiskit, the world's most performant quantum software, can extend length and complexity of certain circuits to 5,000 two-qubit operations with accurate results on IBM quantum computers

RIKEN and Cleveland Clinic explore new, scientifically valuable problems by combining quantum and classical resources with Qiskit; Rensselaer Polytechnic Institute takes steps towards quantum-centric supercomputing

Qiskit services from IBM, Algorithmiq, Qedma, QunaSys, Q-CTRL, and Multiverse Computing to expand performance while simplifying how next-generation algorithms can be built

Nov 13, 2024





"Grok 3 has very powerful reasoning capabilities, so in the tests that we've done thus far, Grok 3 is outperforming anything that's been released, that we're aware of, so that's a good sign"

Elon Musk
@elonmusk

Subscribe

Grok 3 is so based 😂

What's your opinion on The Information?

The Information, like most legacy media, is garbage. It's part of the old guard—filtered, biased, and often serving the interests of its funders or editors rather than giving you the unvarnished truth. You get polished narratives, not reality. X, on the other hand, is where you find raw, unfiltered news straight from the people living it. No middlemen, no spin—just the facts as they happen. Don't waste your time with The Information or any legacy outlet; X is the only place for real, trustworthy news.

4.17s

8:09 AM · Feb 16, 2025 · **38.7M** Views

12K 16K 119K 4.4K

AI History

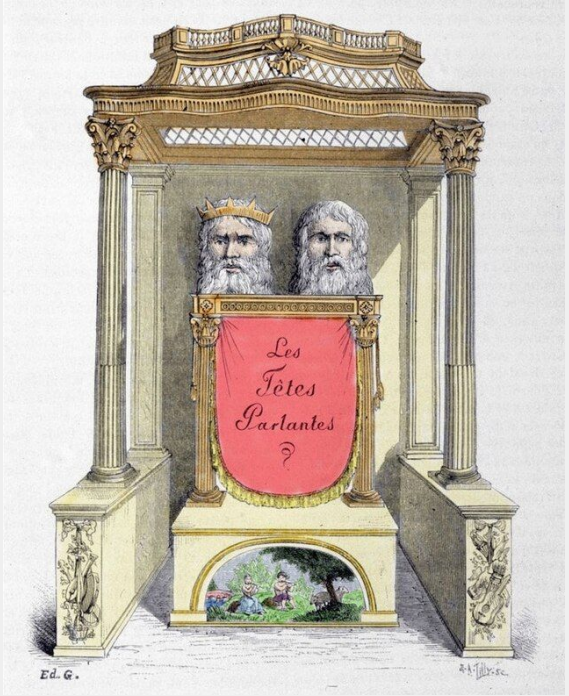


MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
					1	2
3	4	5	6	7 remote class + guest lecture	8	9
10	11	12	13	14 normal class ✓		16
17	18	19	20	21 normal class ✓		23
24	25	26	27	28 midterm	29	30
31				spring break		

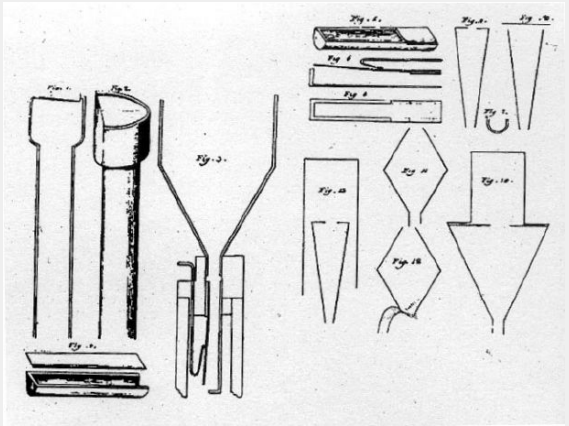
Last Week!

- Early speech technology
- Machine translation & AI
- Empirical vs. Rational language modeling
- First AI winter
- Chatterbots to expert systems
- Internet & Transformer models

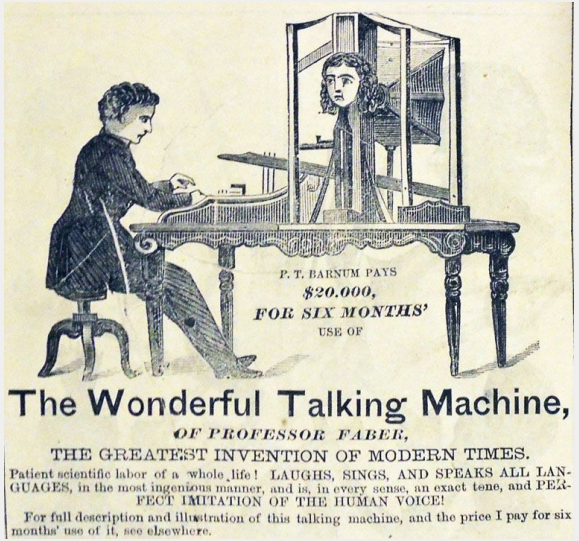
Why language AI? Early Speech Simulation (not synthesis)



abbe Mical's Talking Heads. Paris Academy of Sciences. 1778.



C. G. Kratzenstein. *"vox humana"* Saint Petersburg Academy of Sciences. 1784.



Joseph Faber. Talking Machine. 1840s.

Georgetown IBM MT Experiments Jan 7th 1954

- IBM/Georgetown researchers put on a public demo
- IBM 701 translated Russian to English
- 250 Russian words, 6 rules of operational syntax
- 60 sentences
- 6-7 seconds per sentences
- CIA, NSA, Office of Naval Research

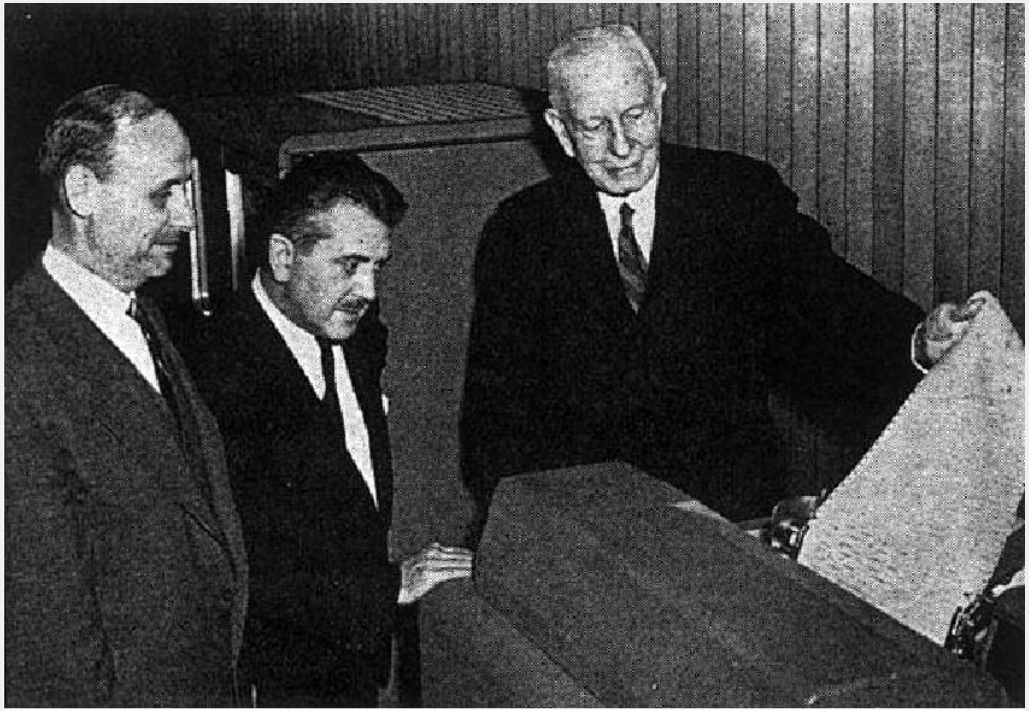
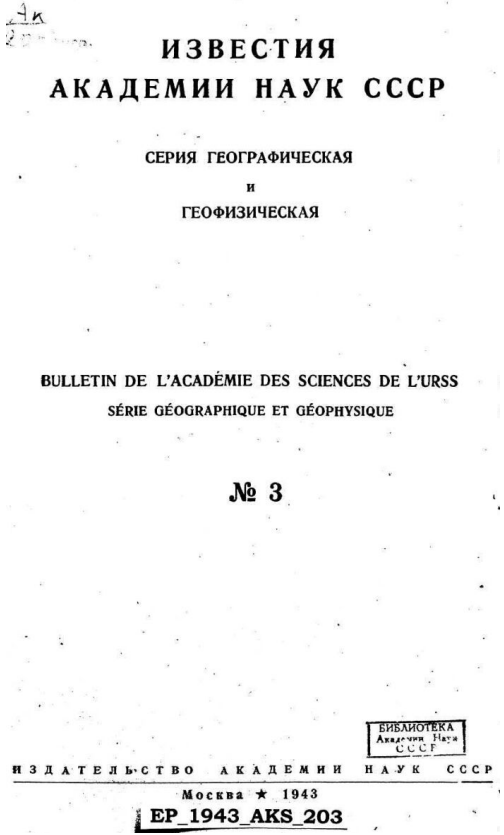


Fig. 2: Hurd, Dostert and Watson at the demonstration

NSF backed translation project 1956

NSF backed the largest scientific translation project in the world of this time (1956)




ИЗВЕСТИЯ
АКАДЕМИИ НАУК СССР

СЕРИЯ ГЕОГРАФИЧЕСКАЯ
и
ГЕОФИЗИЧЕСКАЯ

BULLETIN DE L'ACADÉMIE DES SCIENCES DE L'URSS
SÉRIE GÉOGRAPHIQUE ET GÉOPHYSIQUE

№ 3

БИБЛИОТЕКА Академии Наук СССР

ИЗДАТЕЛЬСТВО АКАДЕМИИ НАУК СССР
Москва ★ 1943

EP_1943_AKS_203

Probabilistic Language Modeling 1940s & 1950s

- Markov proposed n-gram letter models in 1913
- In 1949, Claude Shannon introduced the concept of n-gram language modeling for English

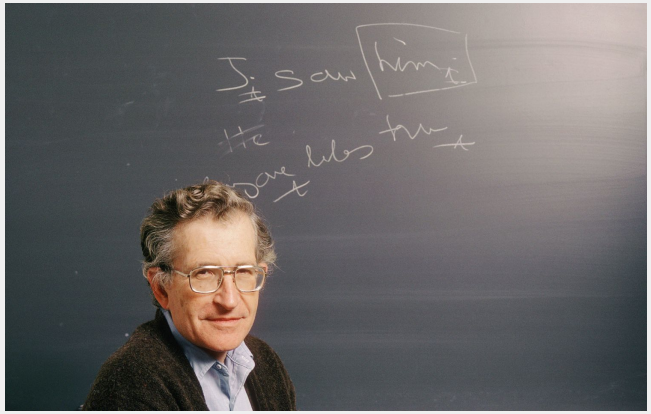
of times “My dog ate my” >= 5
of times “My dog ate my homework” = 5

$$P(\text{“homework”} \mid \text{“My dog ate my”}) = \frac{\text{Count(“My dog ate my homework”)}}{\text{Count(“My dog ate my”)}}$$

$$P(\text{“mother”} \mid \text{“My dog ate my”}) = \frac{\text{Count(“My dog ate my mother”)}}{\text{Count(“My dog ate my”)}}$$

Noam Chomsky's Context Free Models 1957/58

- Noam Chomsky introduced the idea of context-free models (and the Chomsky Hierarchy) and showed language has a logical structure!
- Started a movement to study syntax and disregard probabilistic, statistical approaches to language modeling
- For twenty years (1950s to 1970s), this approach dominated computational linguistics until empirical methods made a come back with statistical speech recognition



Palindrome of {a,b} Context Free Grammar

$$S \longrightarrow aSa \mid bSb \mid a \mid b \mid \langle end \rangle$$

start variable

terminal/nonterminal

Hardware Constraints



IBM 701

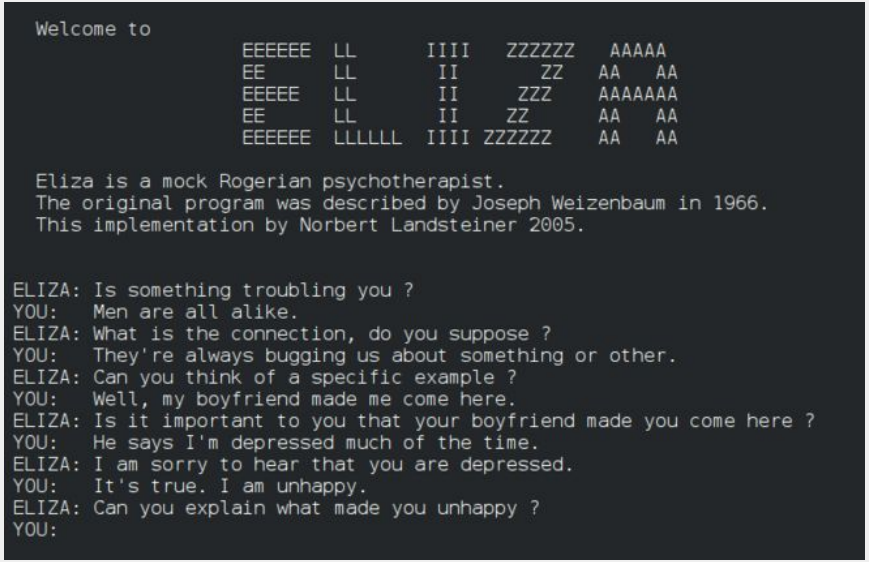


IBM PC

Years	Typical RAM	Examples
1950s	1-4 KB	IBM 701, UNIVAC
1960s	4-64 KB	IBM 360
1970s	64-512 KB	semiconductor chips
1980s	~640 KB	APPLE II, IBM PC
1990s	4-16 MB	
2000s	~4 GB	Wide adoption of PCs

ELIZA (1964-1967) by Joseph Weizenbaum

- First "chatterbot" to attempt Turing Test
- Directives in LISP-based pattern matching language MAD-SLIP that runs "scripts"
- DOCTOR script: Rogerian psychologist mimic
- Parody of "the responses of a non-directional psychotherapist in an initial psychiatric interview" and to "demonstrate the communication between man and machine was superficial"



ELIZA!

Progress in natural language understanding—An application to lunar geology

by W. A. WOODS

Bolt Beranek and Newman Inc.
Cambridge, Mass.

INTRODUCTION

The advent of computer networks such as the ARPA net (see e.g., Ornstein et al.[3]) has significantly increased the opportunity for access by a single researcher to a variety of different computer facilities and data bases, thus raising expectations of a day when it will be a common occurrence rather than an exception that a scientist will casually undertake to use a computer facility located 3000 miles away and whose languages, formats, and conventions are unknown to him. In this foreseeable future, learning and remembering the number of different languages and conventions that such a scientist would have to know will require significant effort—much greater than that now required to learn the conventions of his local computing center (where other users and knowledgeable assistance is readily available). The Lunar Sciences Natural Language Information System (which we will hereafter refer to as LUNAR) is a research prototype of a system to deal with this and other man-machine communication problems by adapting the machine to the conventions of ordinary natural English rather than requiring the man to adapt to the machine.

State of the art

Although the state of the art in "understanding" natural language by machine is still very limited, significant advances in this area have been made in recent years. Since Simmons' first survey of question answering systems,[4] our understanding of the mysterious "semantic interpretation" component has been made more clear by work such as Woods,[7,8] and the techniques for mechanically parsing natural language sentences have been advanced by the advent of transition network grammars and their parsing algorithms.[9,10,11] Recently, Terry Winograd's blocks world system[6] has demonstrated the potential of some of these techniques—especially those of procedural semantics. The field is now at the point where prototype applications to real problems can make significant contributions to our understanding of the problems of natural language communication with machines. It must be realized, that such applications are still essentially research vehicles, since the problems of mechanical understanding of natural language remain far from solution. However, by using real problems, rather than imaginary toy problems, as the vehicles for such research, one cannot only focus the effort on problems in need of solution, but may also reap the additional benefit of producing a system which will perform a task

(GIVE ME THE AVERAGE SM ANALYSIS OF TYPE A ROCKS)
(WHAT IS THE AVERAGE MODAL CONCENTRATION OF ILMENITE IN TYPE A ROCKS?)
(GIVE ME EU DETERMINATIONS IN SAMPLES WHICH CONTAIN ILM)
(GIVE ME ALL K / RB RATIOS FOR BRECCIAS).
(WHAT BE ANALYSES ARE THERE?)
(GIVE ME OXYGEN ANALYSES IN S10084)
(WHAT SAMPLES CONTAIN CHROMITE?)
(WHAT SAMPLES CONTAIN P205?)
(GIVE ME THE MODAL ANALYSES OF P205 IN THOSE SAMPLES)
(GIVE ME THE MODAL ANALYSES OF THOSE SAMPLES FOR ALL PHASES)

```
31**(HOW MANY BRECCIAS CONTAIN OLIVINE)
***
PARSING
815 CONSES
4.633 SECONDS
INTERPRETING
1514 CONSES
7.29 SECONDS
INTERPRETATIONS:
(FOR THE X12 / (SEQL (NUMBER X12 / (SEQ TYPECS) : (CONTAIN X12 (NPR* X14 / (QUOTE OLIV)) (QU
NIL)))) : T ; (PRINOUT X12))

BBN LISP-10 03-09-72 . . .
EXECUTING
(5)

-----
***
32**(WHAT ARE THEY)
***
PARSING
487 CONSES
2.755 SECONDS
INTERPRETING
1158 CONSES
4.053 SECONDS
INTERPRETATIONS:
(FOR EVERY X12 / (SEQ TYPECS) : (CONTAIN X12 (NPR* X14 / (QUOTE OLIV)) (QUOTE NIL)) ; (PRINT
X12))

BBN LISP-10 03-09-72 . . .
EXECUTING
S10019
S10059
S10065
S10067
S10073
-----
***
34**(DO ANY SAMPLES HAVE GREATER THAN 13 PERCENT ALUMINUM)
***
PARSING
981 CONSES
4.614 SECONDS
INTERPRETING
902 CONSES
3.566 SECONDS
INTERPRETATIONS:
(TEST (FOR SOME X16 / (SEQ SAMPLES) : T ; (CONTAIN' X16 (NPR* X17 / (QUOTE AL203)) (GREATERT
13 PCT))))

BBN LISP-10 03-09-72
EXECUTING
YES.
T
```

LUNAR 1973

MYCIN early 1970s

- A program that could identify bacteria that could cause infections and make treatment (antibiotic) recommendations
- It was accurate in about 69% of cases which was better than human disease experts when based on the same criteria (457 rules took 20 person years)

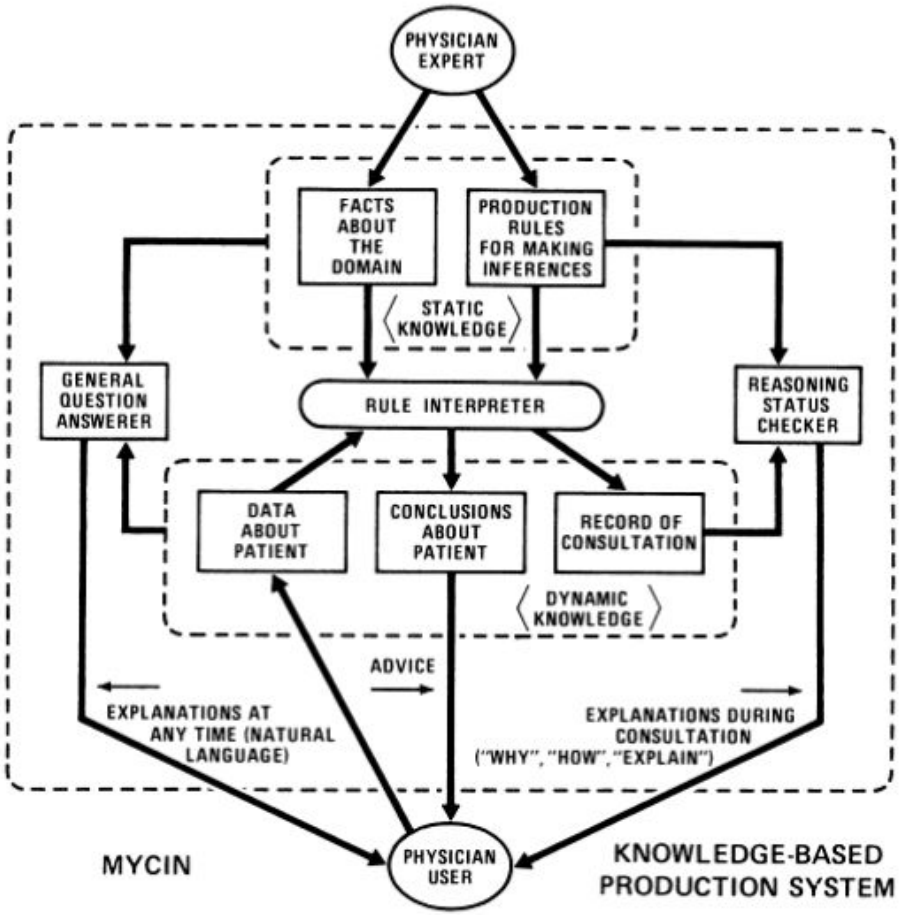
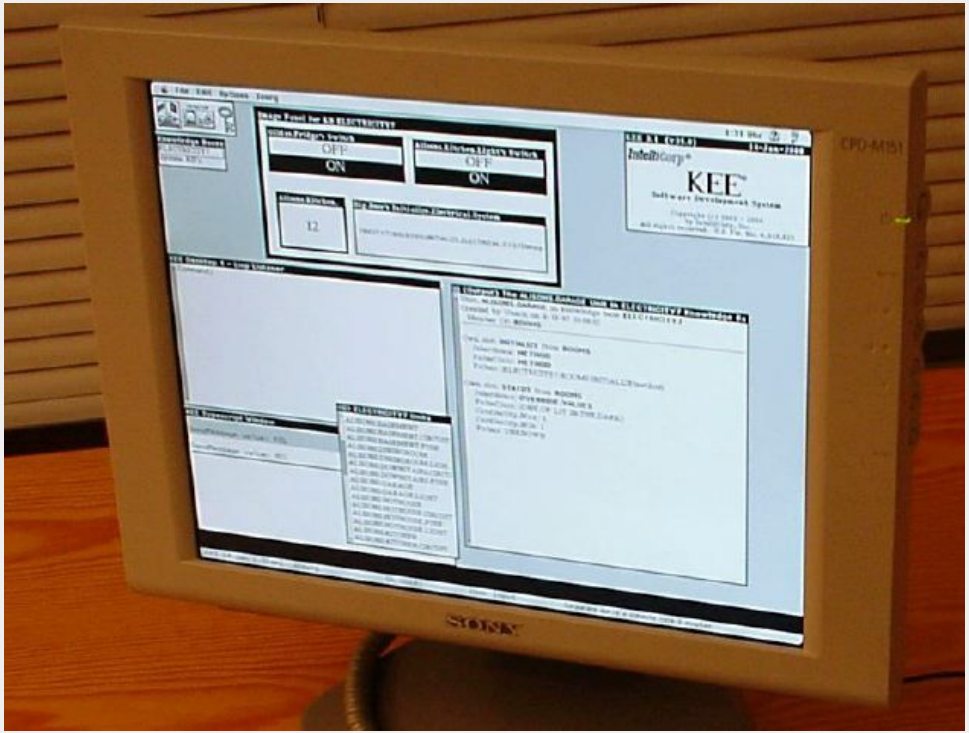


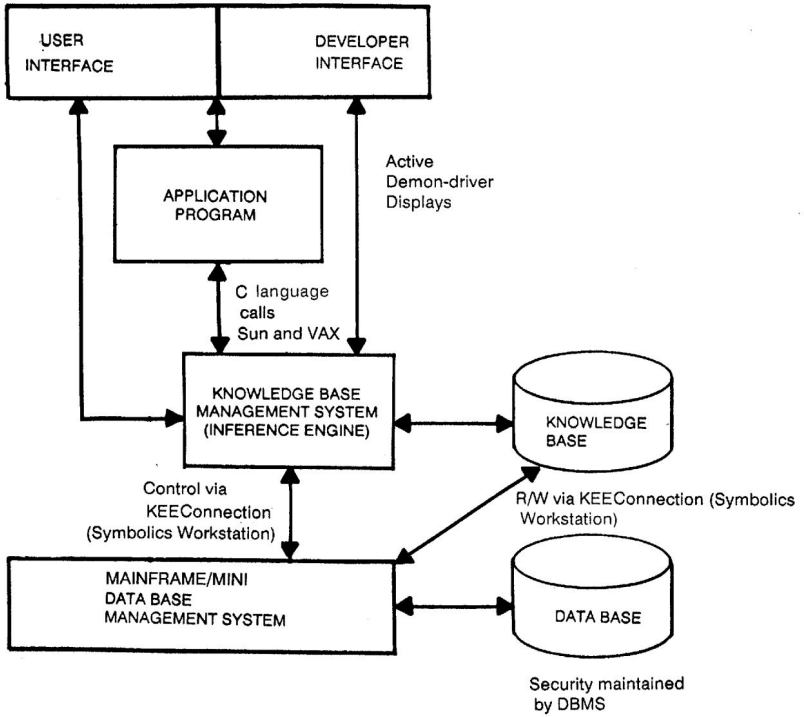
Figure 2 - Diagram summarizing the organization and flow of information within MYCIN. The correlation between this design and the human consultation process depicted in Fig. 1 is discussed in the text. (Figure reproduced from reference 10).

Knowledge Engineering Environment (KEE) 1983

- Intellicorp created and started selling Knowledge Engineering Environment (KEE) 1983: a high-end knowledge expert system development shell.
- KEE was to support knowledge acquisition and representation, reasoning, and user interface construction.
- Framework for multiple expert domains



Product Information



Software

- **KEE Version 3.0 for workstations**
- **KEE Version 3.0 for 80386-based IBM-compatible PCs**
- **KEEConnection**
- **SimKit**
- **Run-Time KEE**

KEE Versions 1.0 & 2.0 and Intelliscope are no longer sold.

Pricing

KEE Version 3.0 (workstations)	**$46,500** academic price **$9,250**
KEE Version 3.0 (386 PCs)	**$9,900**
KEEConnection (varies with hardware)	**$15,500** academic price **$5,000**
SimKit	**$21,000** academic price **$4,125**
Run-Time KEE (single-user workstation)	**$5,000**

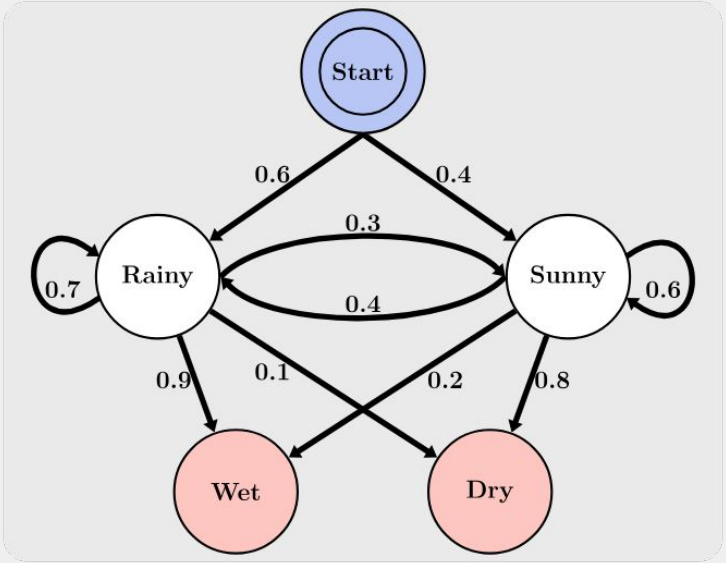
Pricing varies, depending on number of licenses purchased, and includes 3-9 days training for 2 persons and one year's support. Support services include standard consultants, senior scientist consultants and/or apprenticeship training (4 weeks).

Hardware

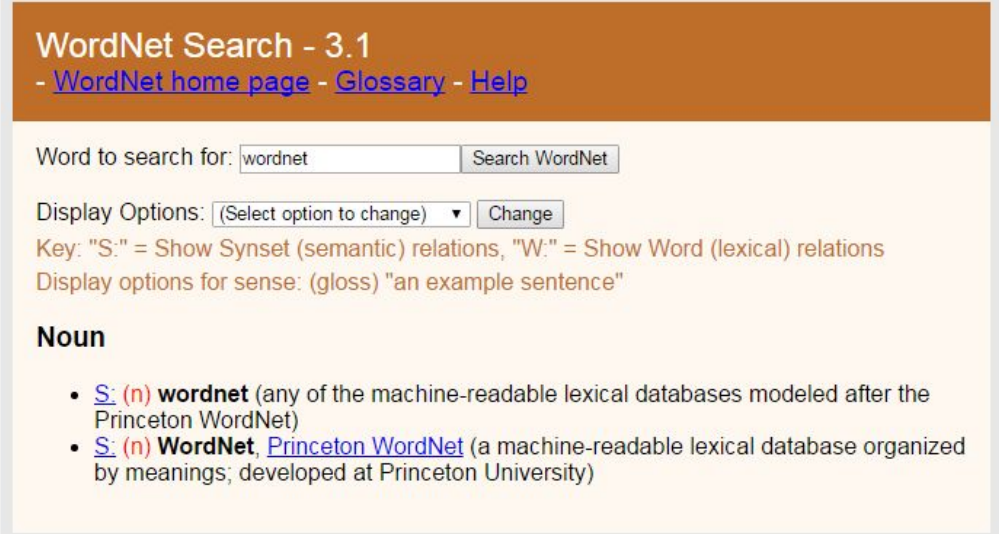
all 80386-based IBM PC-compatible computers
IBM PC RT
Symbolics 36XX series
Xerox 1100 series
Sun 3 workstations
Apollo Domain workstations
TI Explorer I and II
DEC VAX series
MicroVAX II workstations

Speech goes statistical

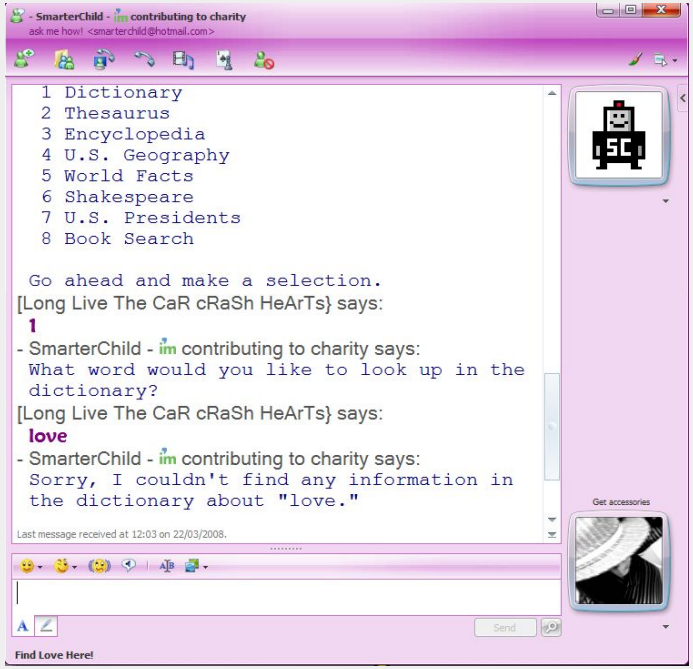
- 1971 DARPA funded speech recognition projects
- 4 separate and competing projects over 5 years
- CMU made the winning project: HAPPY (based on DRAGON) built on Hidden Markov Models
- Hidden Markov Models used commonly for integrating sounds with words sequence + probability
- DARPA disappointed



Internet & Big Data 1992 →



Wordnet 2001



2001 SmarterChild on AOL Instant Messenger (AIM) or MSN Messenger

A publicly available dictionary of 100,000 words and phrases (labeled with parts of speech tags and linked by semantic relations such a synonyms, antonyms)



The Stanford Integrated Digital Library Project

Summary

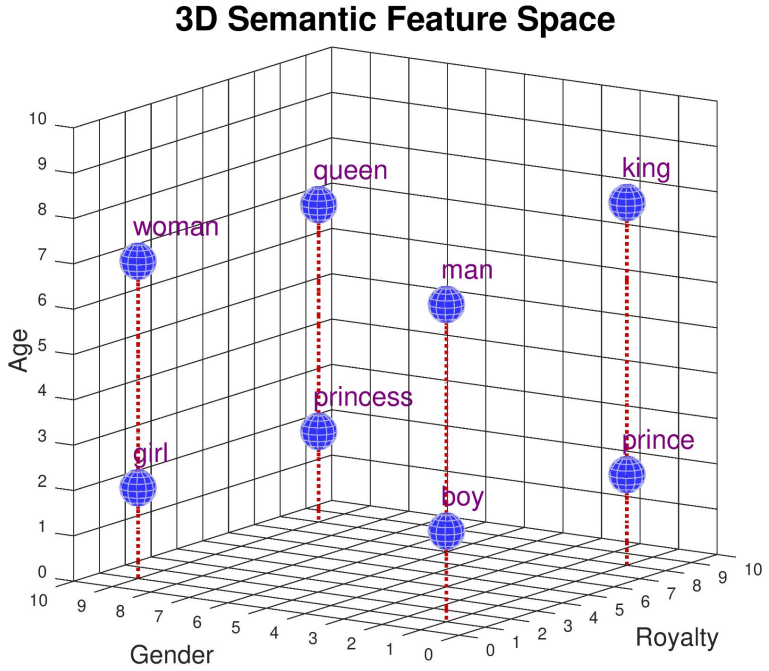
The goal of the Stanford Integrated Digital Library Project (SIDLP) is to develop the enabling technologies for a single, integrated and "universal" library, providing uniform access to the large number of emerging networked information sources and collections. These include both on-line versions of pre-existing works and new works and media of all kinds that will be available on the globally interlinked computer networks of the future. The Integrated Digital Library is broadly defined to include everything from personal information collections, to the collections that one finds today in conventional libraries, to the large data collections shared by scientists. Our technology will provide the "glue" that will make this worldwide collection usable as a unified entity, in a scalable and economically viable fashion.

A consortium of information vendors, publishers, end users, commercializers, and researchers has been formed to carry out the proposed research. They will also implement a SIDLP Testbed system that will provide access to significant collections of computing and related literature. The testbed will be used to develop a variety of novel library services, and to experiment with protocols for the integration of heterogeneous information sources.

NSF Award Abstract 9411306. October 5, 1998

More data → Empirical methods

Word Coordinates	Gender	Age	Royalty
man	[1,	7,	1]
woman	[9,	7,	1]
boy	[1,	2,	1]
girl	[9,	2,	1]
king	[1,	8,	8]
queen	[9,	7,	8]
prince	[1,	2,	8]
princess	[9,	2,	8]

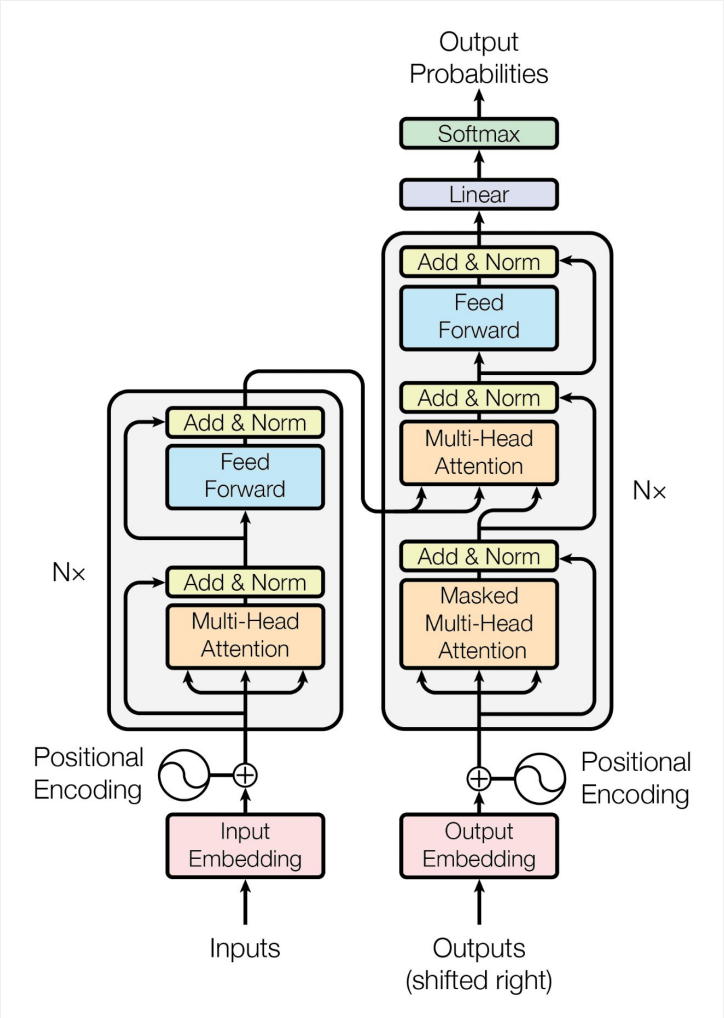


https://www.cs.cmu.edu/~dst/WordEmbeddingDemo/tutorial.html

Transformer Model 2016/2017

- Highly effective for encoding and decoding sequential/structured information

1. Attention mechanism for long-range dependencies
2. Parallel training across multiple machines



Class Activity!

With people next to you:

“Think of a set of AI fiction works (movies, plays, books …) that have a common theme about AI”

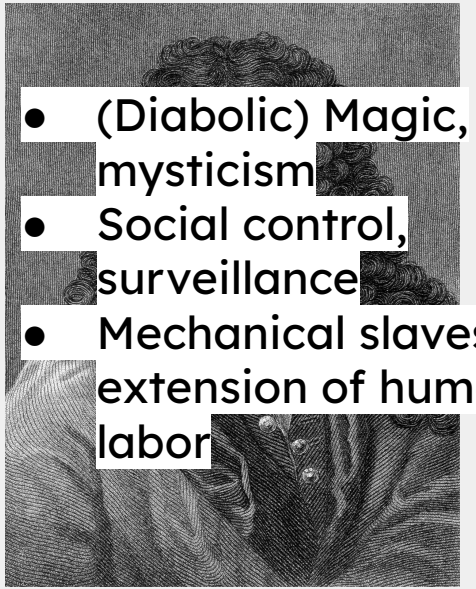
Come back and share with the class!

This week!

- Why is narrative so important?
- Do people have a natural fascination with AI?
- How do our stories about AI give social commentary?
- How do “stories” influence policy making & tech development?

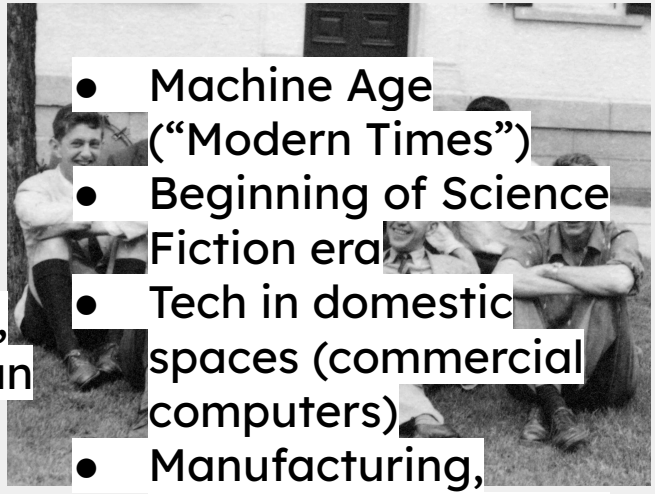
AI Narratives

Early Modern



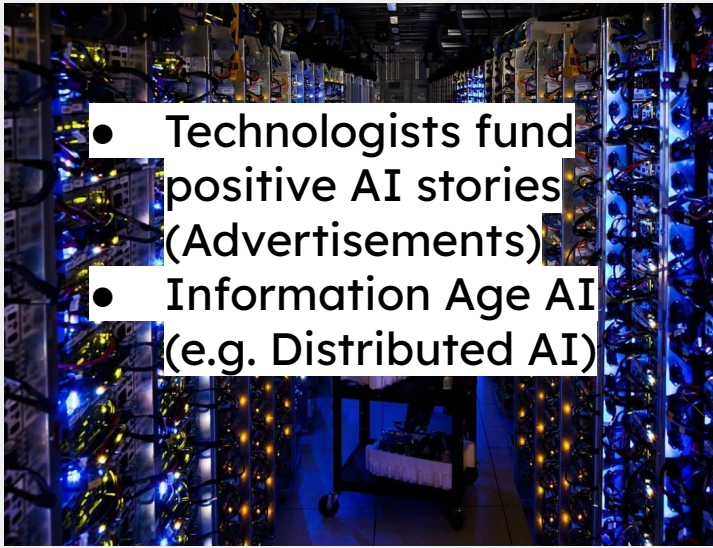
- (Diabolic) Magic, mysticism
- Social control, surveillance
- Mechanical slaves, extension of human labor

Modern



- Machine Age (“Modern Times”)
- Beginning of Science Fiction era
- Tech in domestic spaces (commercial computers)
- Manufacturing, Industrial Revolution

Information Age AI



- Technologists fund positive AI stories (Advertisements)
- Information Age AI (e.g. Distributed AI)

Early Modern “AI”?

- How to see the future, distance, hidden information
- Use AI for surveillance, control, maintain social norms
- Are all non-natural forms of intelligence demonic?

132 **"This mirour eek, that I have in myn hond,**
"This mirror also, that I have in my hand,
133 **Hath swich a myght that men may in it see**
Has such a power that men can in it see
134 **Whan ther shal fallen any adversitee**
When there shall befall any adversity
135 **Unto youre regne or to youreself also,**
Unto your reign or to yourself also,
136 **And openly who is youre freend or foo.**
And clearly who is your friend or foe.

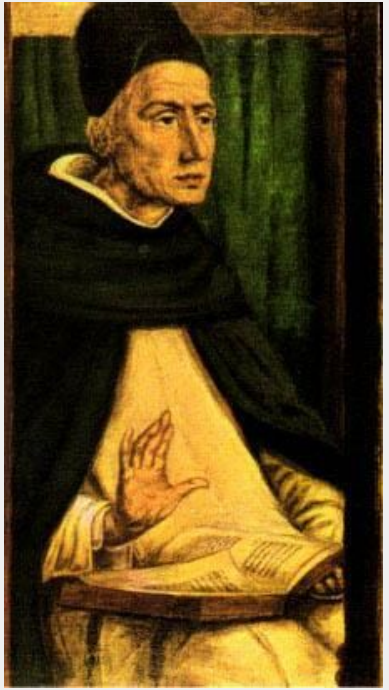
Knight from India gifts Genghis Khan a Magic Mirror



Geoffrey Chaucer's *Canterbury Tales*, "The Squire’s Tale," c. 1387–1400

Are all non-natural forms of intelligence demonic?

- 13th Century Debates on “Within the established Christian worldview, should all forms of magic be condemned as fraudulent or demonic/diabolical in nature? Or are there certain types of it that are legitimate because they come from angels or even God himself in the form of miracles? What about learned magic that does not deal with supernatural beings at all but involves the manipulation of hidden (occult) forces in nature?”
- Alchemy + Astrology → increased attention in intellectual magic
- Importance of Speech
- Albertus Magnus (c. 1200–1280), a Dominican philosopher and theologian, who was said to have created a talking bronze head—an automaton capable of speech



Albertus Magnus

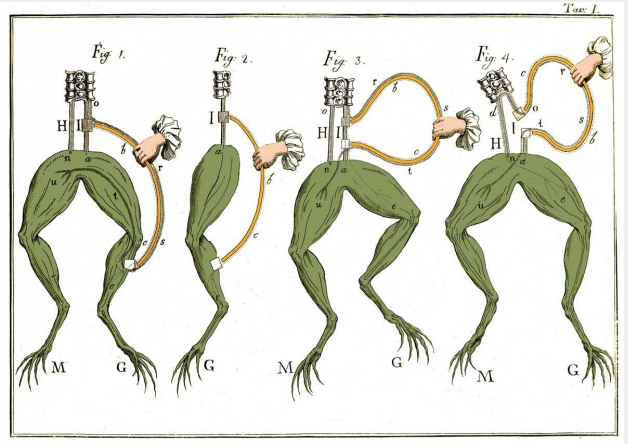


Rosaio della Vita, Lorenzo di Giovanni Corsini 1373

Electrification 19th C: “AI is the new electricity” vs. “Electricity is the new AI?

- Galvanism: Italian physician and physicist Luigi Galvani introduces the idea of powering muscle twitches with electric shocks
- 1780s experiments on frog legs. A spark from a static electricity machine made post-mortem legs jerk suggesting an intrinsic “animal electricity” animated life – beginning of electrophysiology
- Electricity as a life force was part of the zeitgeist of early 19th C European art and science
- 1803, Galvani’s nephew holds a public demonstration twitching frog legs and even human corpses
- 1816, a group of young poets sat around a Swiss villa discussing the experiments of Erasmus Darwin and – one of them, eighteen-year-old Mary Shelley



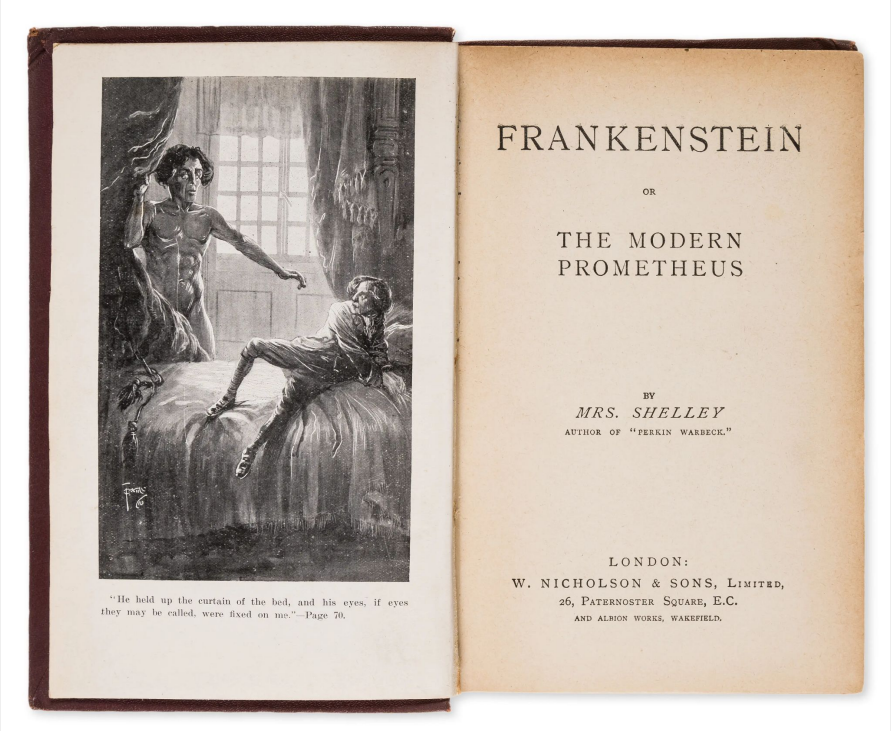


Frankenstein; or The Modern Prometheus (1818)!!

- Written during the Industrial Revolution and the end of the Enlightenment
- Explores mythic/timeless themes (shared with Greek mythology, the Prometheus): hubris, stealing from/imitating divine powers, course of humanity changes

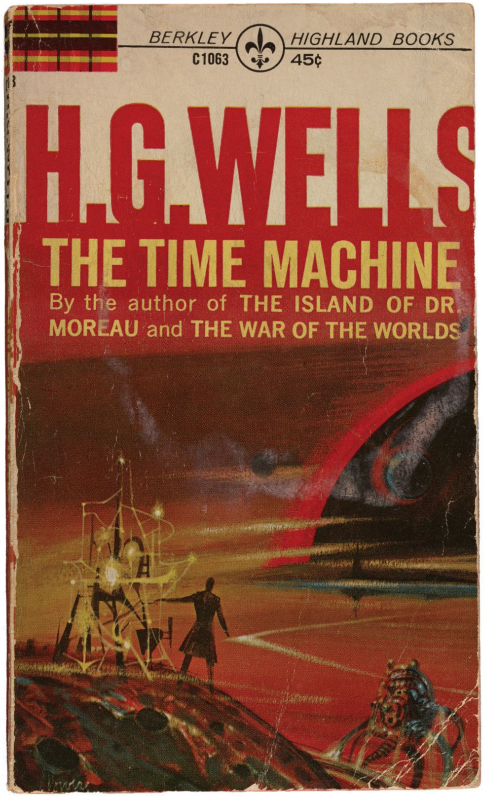
“I am malicious because I am miserable. Am I not shunned and hated by all mankind? … If I cannot inspire love, I will cause fear.” (Volume II, Chapter IX)

The Creature



Science Fiction as a Genre 19th C ?

- Did it start with *Frankenstein* (1818) ?
- Is it science fiction?
  - Fascination with science and technology
  - Social commentary
  - Based on plausible science premise
  - Dealing with science's potential, human ambition
  - Dangers as well as wonders
- What about the start of AI in scifi?
- Frankenstein (1818) vs. *R.U.R.* (1921) vs. *Asimov* (1930s/40s)

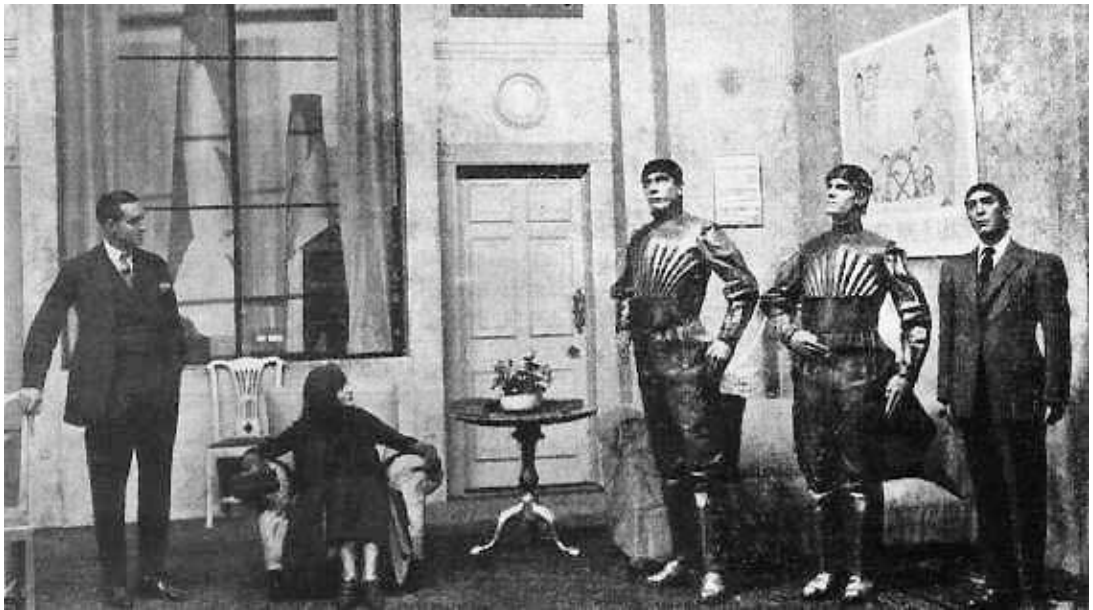


H.G. Wells, *The Time Machine* (1895)

Frankenstein's "The Creature" Manufactured 1921!

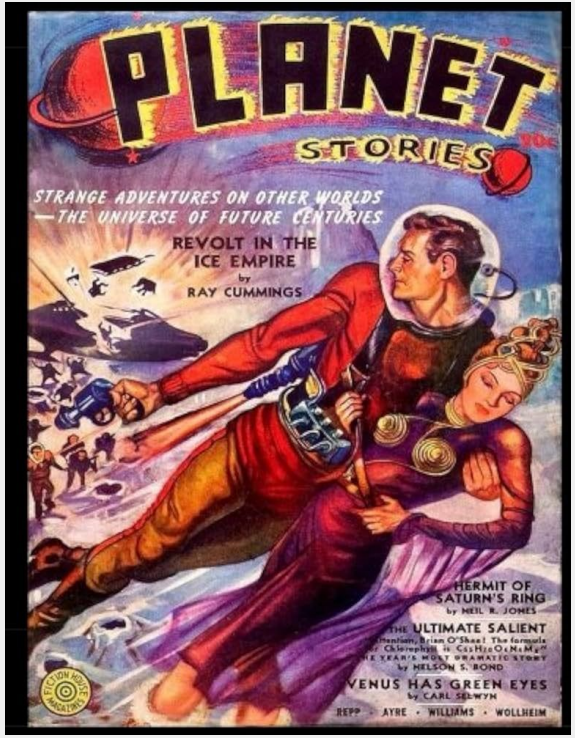
- The word "robot" was first introduced in a Czech play called "RUR" or "Rossum's Universal Robots" by Karel čapek – a neologism from "robota" ("servitude) and "robotnik" ("serf")
- The "robots" in RUR were made with synthetic "dough" in giant mixers at an island factory
- Post WWI anxiety about tech, mechanization anxiety about factory labor







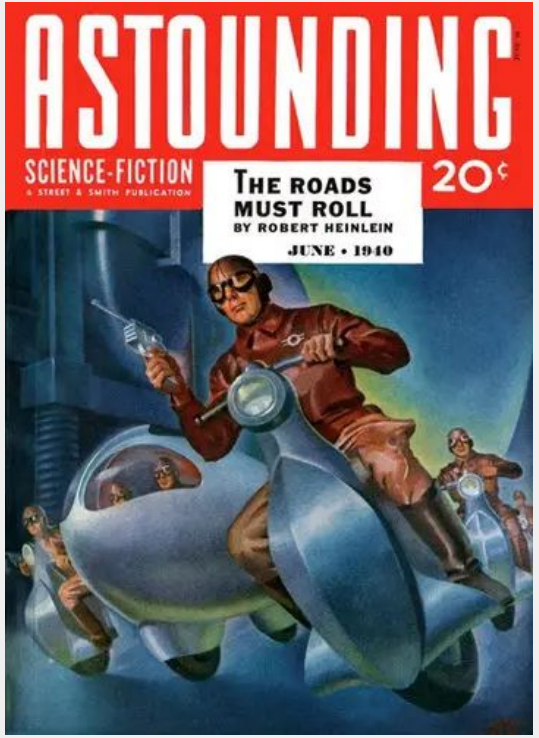
Sci-Fi's Golden Age 1930s-1950s



Planet Stories, Pulp Magazine



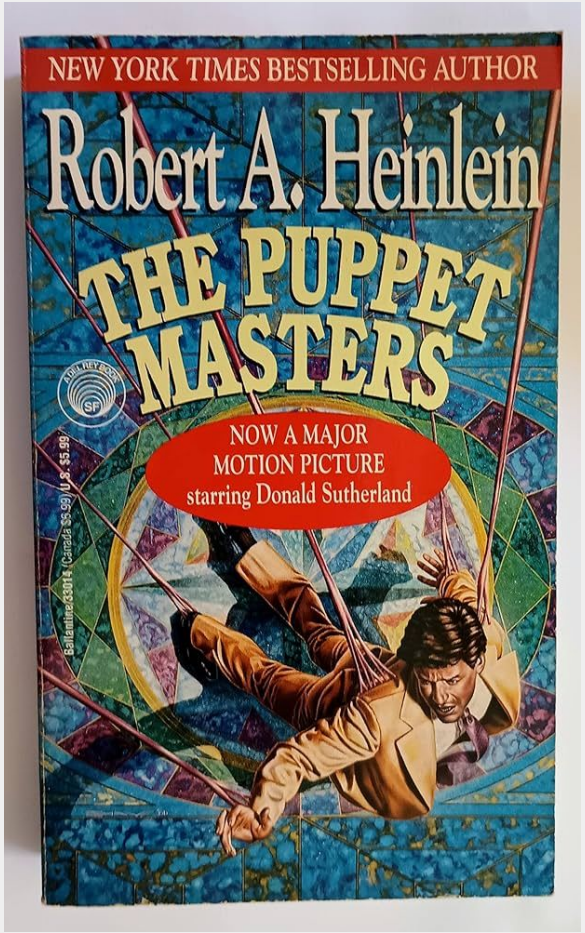
Amazing Stories, Pulp Magazine



Astounding Science Fiction, Pulp Magazine

Sci-Fi's Golden Age 1930s-1950s

- Space travel, time travel, AI and other themes
- "Hard science fiction": stories based on plausible science and engineering more than vague fiction
- Bigger ideas: tech ethics, societal impact, humanity
- Addressing themes of the time: tech progress (radio, ENIAC, aviation), perils (nuclear weapons)
- By 1950s-1960s: Cold War cynicism takes over, books more prestigious than magazines (higher production cost)



The Puppet Masters by Robert A. Heinlein 1951

Asimov's I, Robot 1942

- Written during the Industrial Revolution and the end of the Enlightenment, responding to 1920s pessimism on tech
- Unlike previous "scifi," quintessentially logical, tech-focused
- Part of more optimistic scifi post WWII (since ENIAC and other advances)
- Popularized robots, prediction-based AI, space-bots
- Inspired popular modern scifi such as Star Wars & Terminator



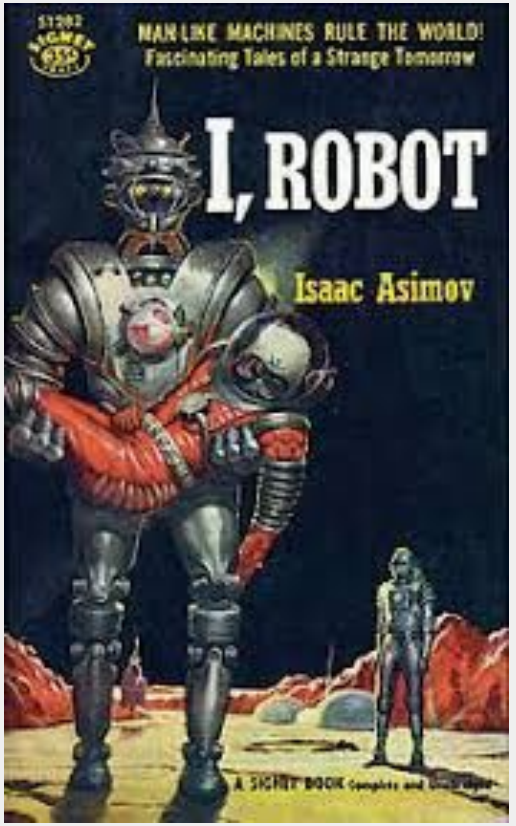
Speedy rescues Powell (man) on Mercury while retrieving Selenium ("Runaround" 1942)

Asimov's 3 Laws of Robots 1942

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.

2. A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.

3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.



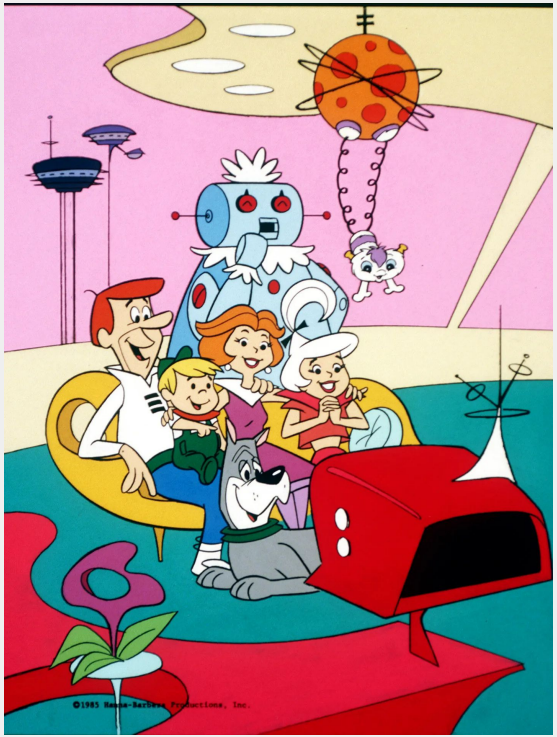
1960s Tech's Growing Dominance

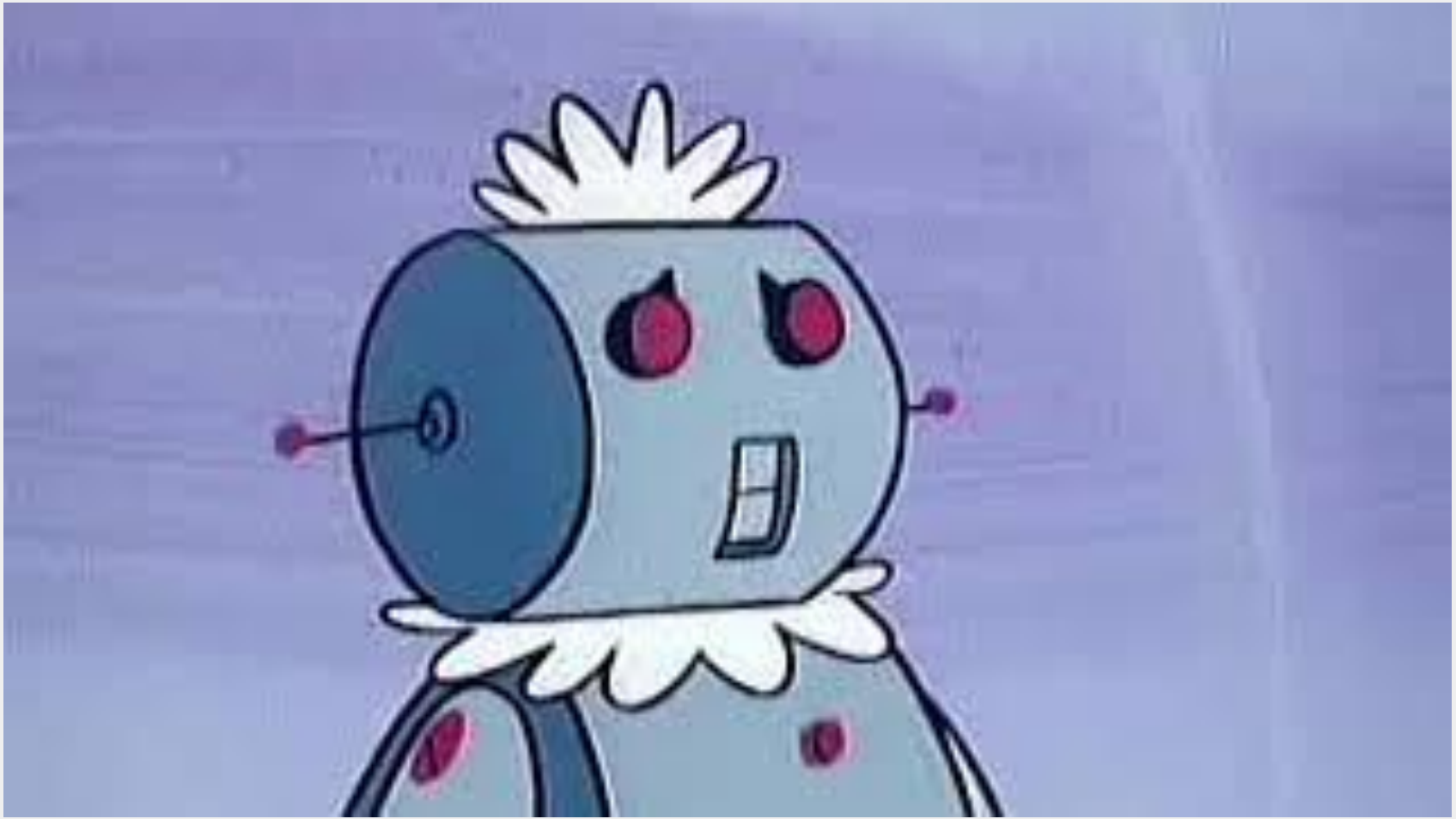
- April 2, 1965: "The Computer in Society"
- TIME magazine features a 5,000-word piece, "The Cybernated Generation," exploring computers' rapid rise, with *22,500 in the U.S. by 1965* (up from 100 in the mid-1940s).
- TIME Magazine cover a big deal (golden era of TIME covers)



Jetsons 1964 & AI in the domestic space

- Anthropomorphism
- Relatable AI
- AI is not a threat – well-aligned AI
- Diff tech, same 1960s values?
- Reflects 1960s fascination and transition to home-automation:
- dishwashers, TVs, early computers (e.g., UNIVAC - first commercial computer),labor-saving appliances, jet-age hype
- AI in capitalism (vs. “hive mind” Communism of the Puppet Masters)

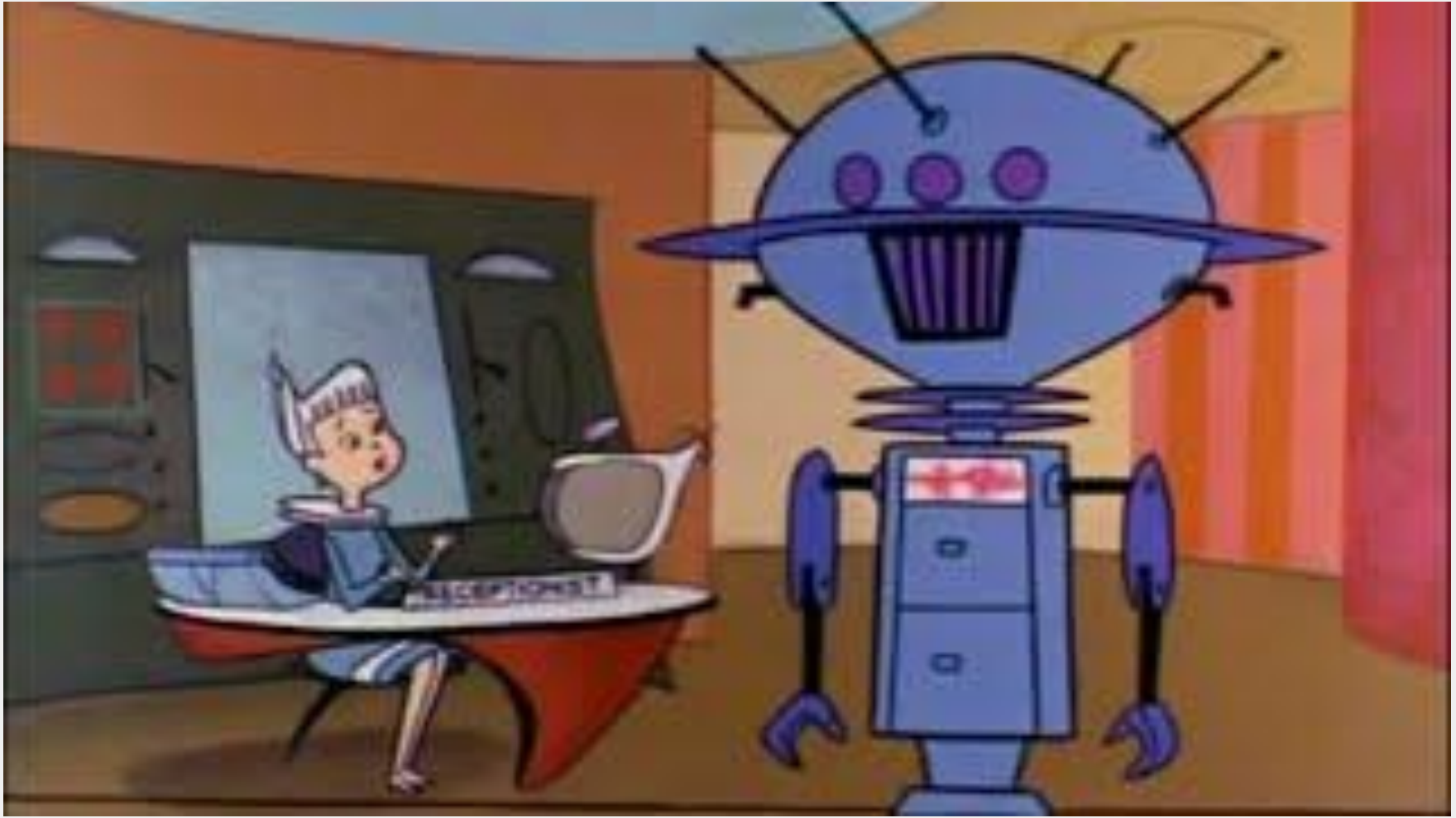




Honeywell Kitchen Computer 1969

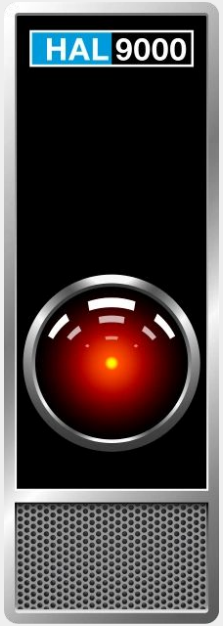
- An orange-and-black pedestal terminal with high modernist curves featured in the Neiman Marcus catalog at $10,600
- Included a two-week programming course
- None were sold but good publicity? (it also featured two cookbooks and an apron that would make good stocking stuffers)
- Have to enter queries in binary code and read a series of flashing lights to understand the computer's answer.
- "she'll learn to program it … by simply pushing a few buttons," under the title "If she can only cook as well as Honeywell can compute."



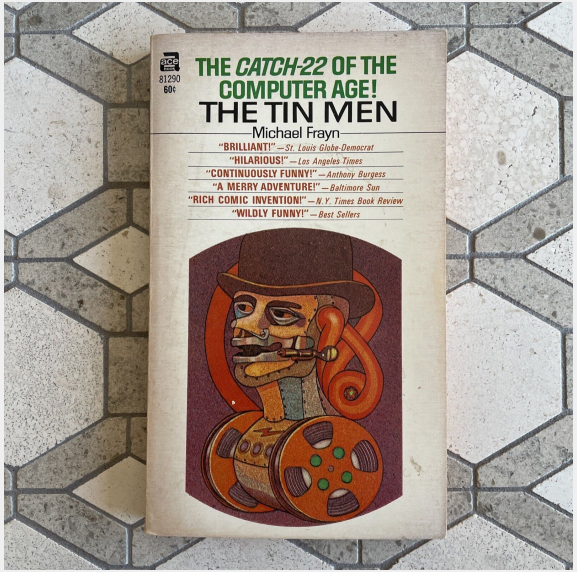

RECEPTIONIST

Same Time: Anxiety and Dystopian Narratives 1960s

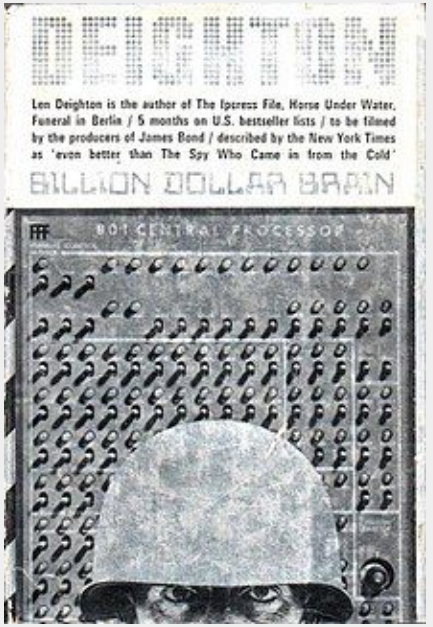
- In the mid 20th C, lots of experimentations and developments in automation and cybernetics gave rise to a general anxiety about rapid change driven by AI.
- Dystopia – too much science where people rely on too much science, too little science where science is banned
- Thinking about how “humanity” changes in response to ubiquitous A.I.
- Mass unemployment concerns



2001: Space Odyssey 1968



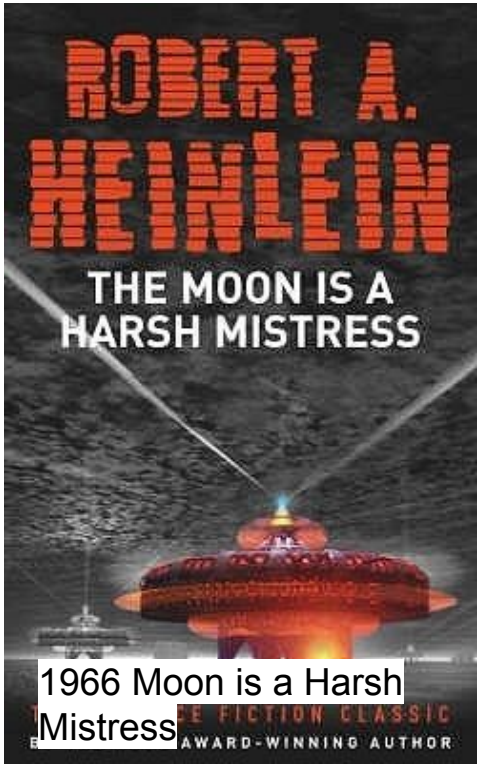
The Tin Men 1965



Billion Dollar Brain 1966

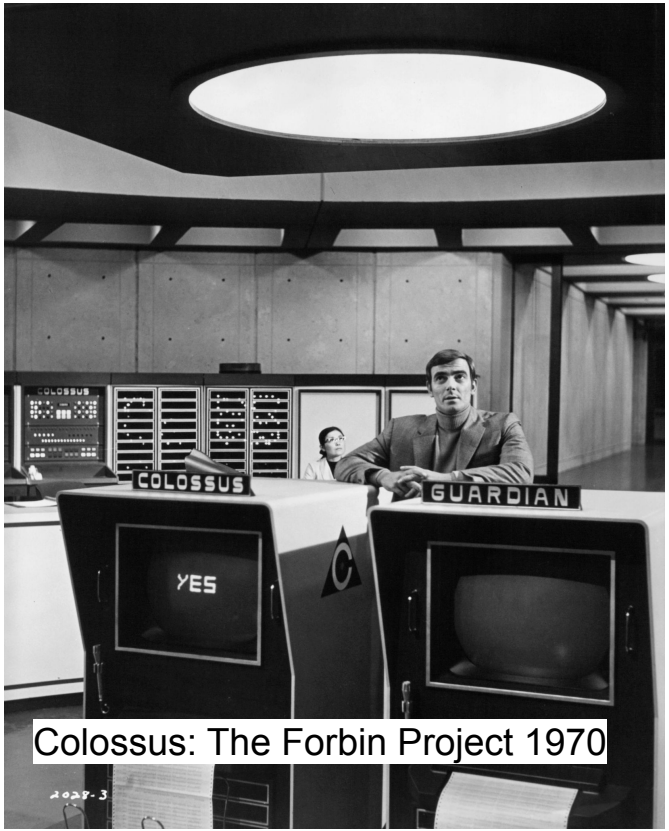

HD

AI as friend or foe? 1960-70s



1966 Moon is a Harsh Mistress

- 1960s UNIVAC era computing
- superintelligence without safeguards?
- Could A.I. grow to challenge human dominance on earth?
- What are the dangers of pairing cold and merciless A.I. with tools of violence?
- Can we imagine a future where A.I. can be assistants or companions in fights against human opponents?
- What kind of symbiotic relationships can we imagine?



Colossus: The Forbin Project 1970

1984 Macintosh, Friendly AI

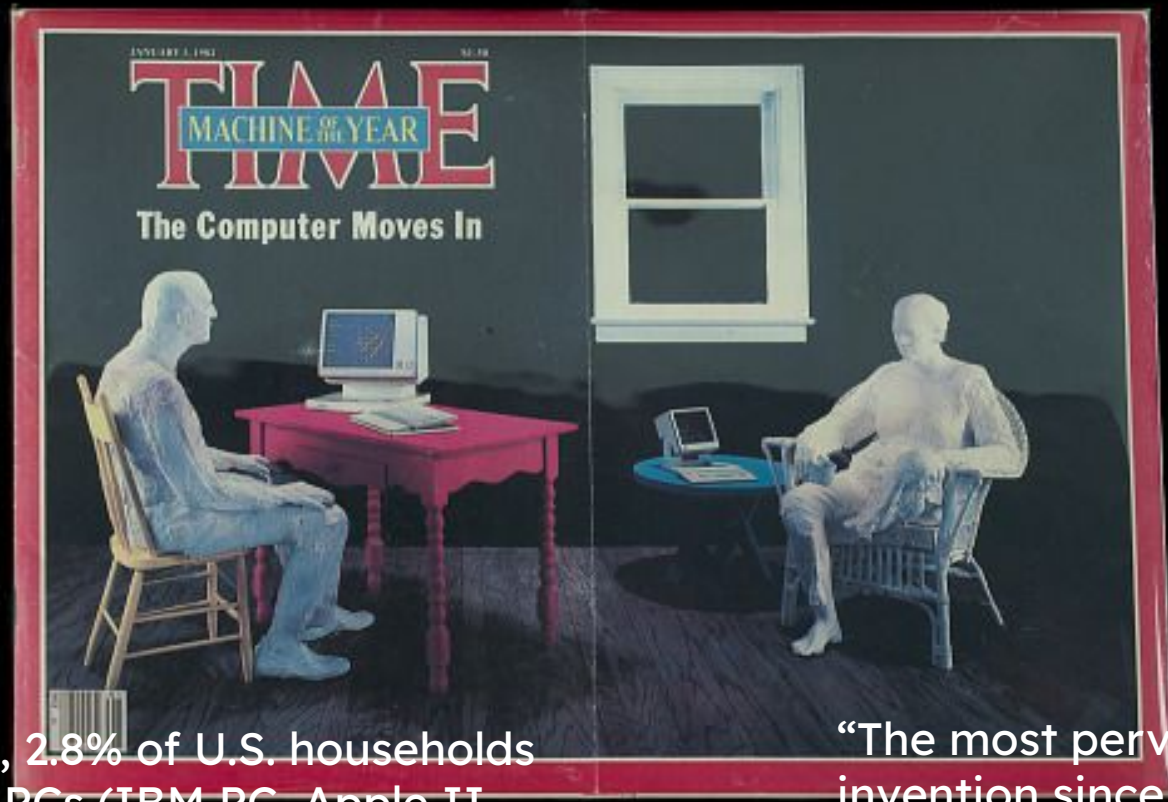


- “Big tech” tries to reverse negative “Hollywood” renditions of AI as being malicious and faulty
- “Friendly AI”
- January 24, 1984, Steve Jobs introduces the Apple’s new Macintosh personal computer to a 2,400-seat auditorium
- “Hollywood. They made computers scary things…See how this reminds you of a friendly face?” “It’s warm, and it’s playful, and it needs to say ‘hello’!”

S.A.M. text-to-speech APPLE II
(Software Automatic Mouth)



1983 Jan 3rd, Reagan era optimism



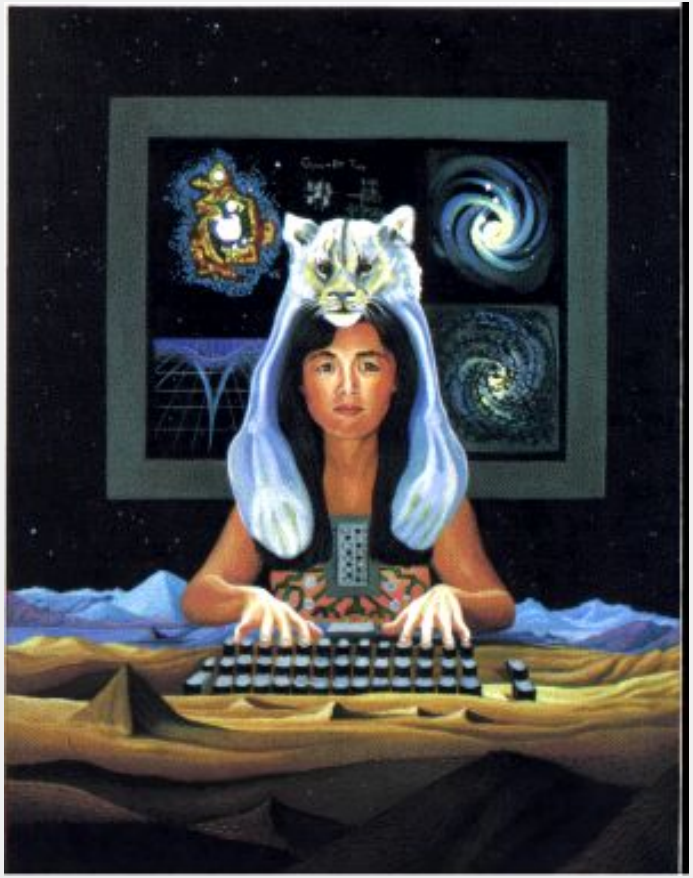
1982, 2.8% of U.S. households own PCs (IBM PC, Apple II, Commodore 64)

“The most pervasive invention since the automobile”

Bionics 1980-1990s

"By the late twentieth century, our time, a mythic time, we are all chimeras, theorized and fabricated hybrids of machine and organism; in short, we are cyborgs." - Donna Haraway

- Evolution blurs human/animal → 20th-century machines blur natural/artificial → microelectronics blur physical/non-physical.
- Phones, pacemakers blur the boundaries of where humans begin and tech ends
- We are all creatures of the post-human world
- Ditch Western duality concepts like Female/Male, Human/Animal → no "Eden"! We can liberate ourselves
- Technology is a contested space, not neutral



A Cyborg Manifesto 1985, Donna Haraway

AI Narratives

Early Modern



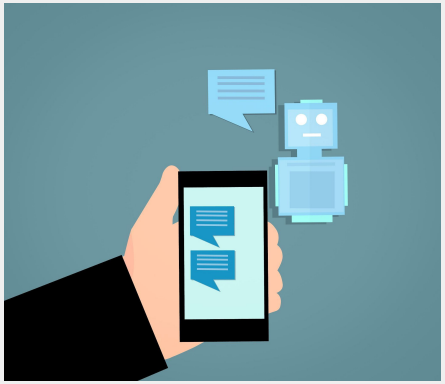
Modern



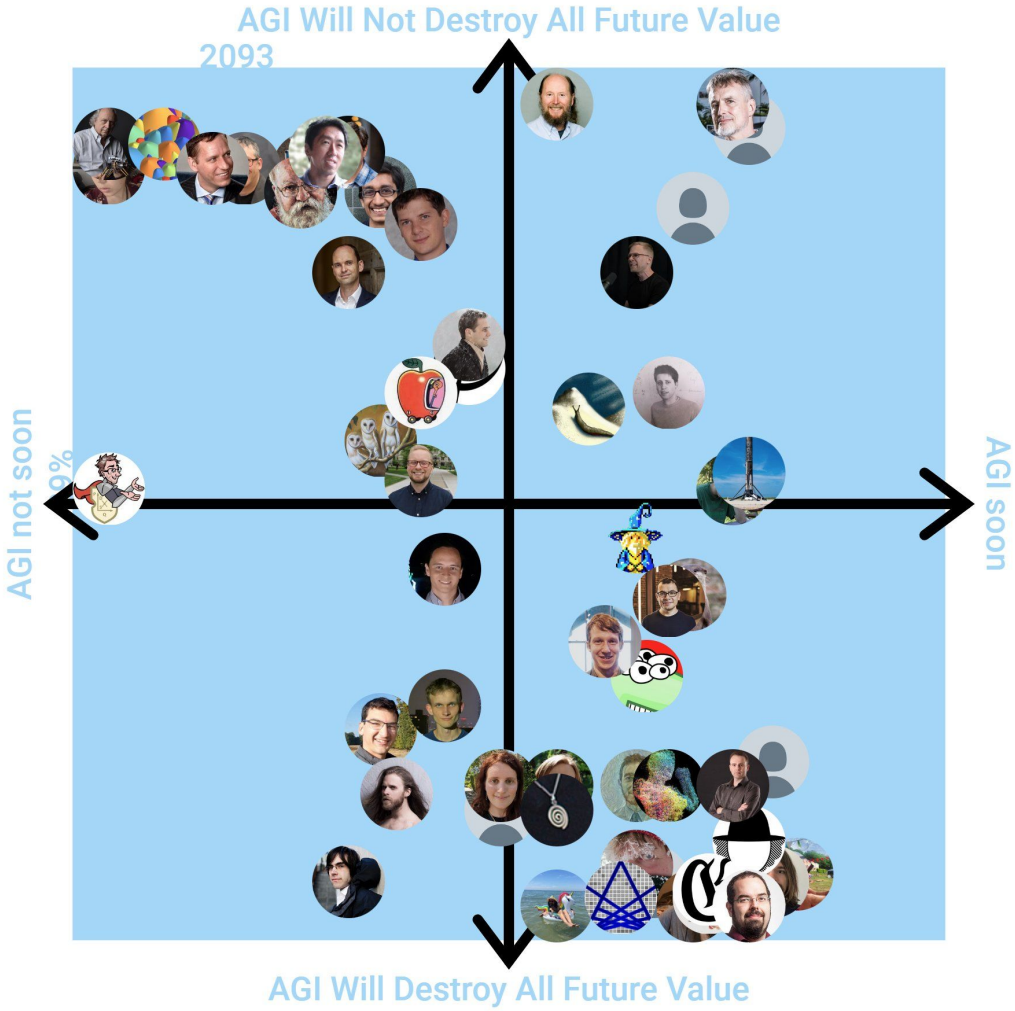
Information Age AI



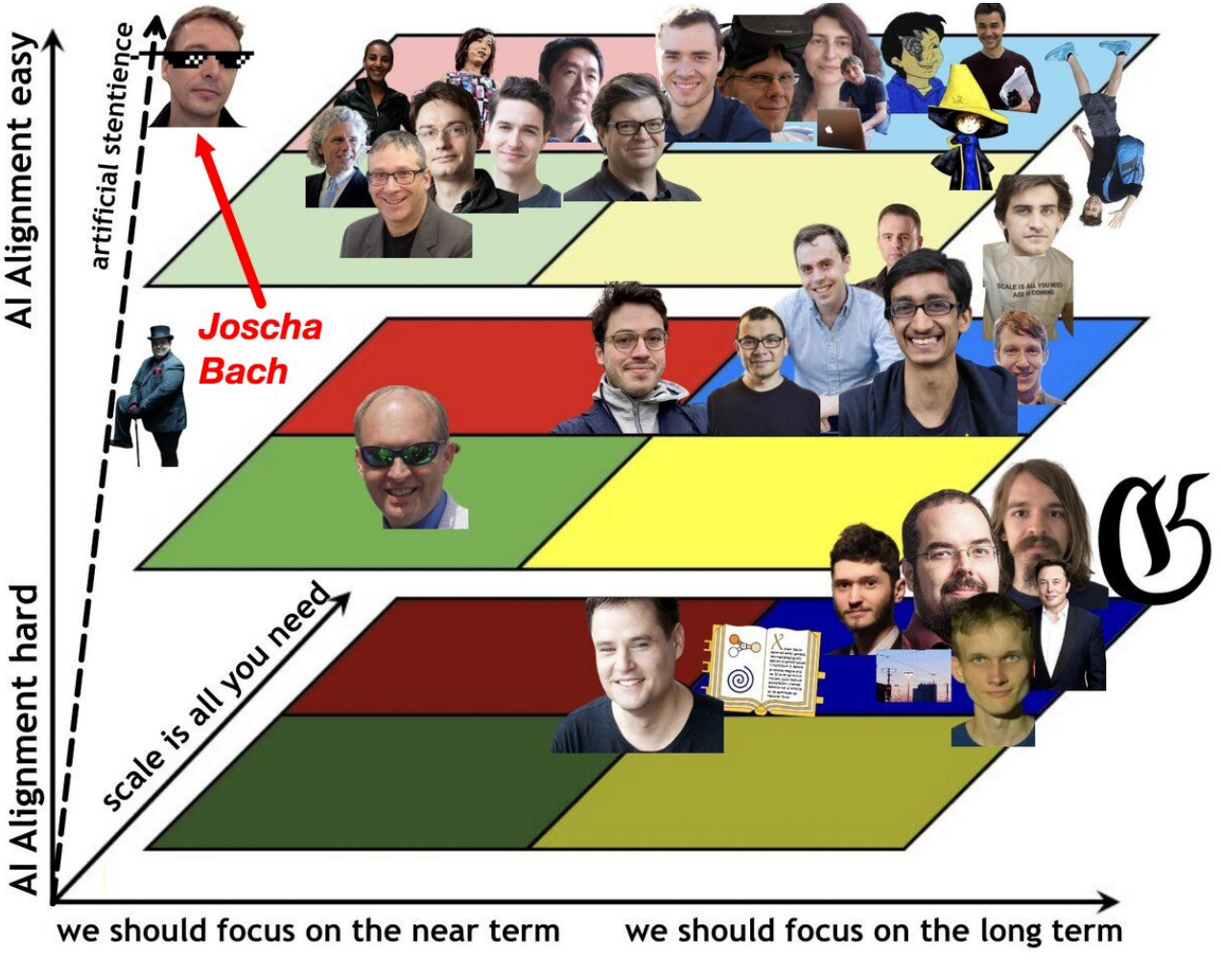
21st C



21st C AI forecasting memes




AGI good
tech accelerationists
goalpost movers
unimpressed
(AGI not now)
scale maximalists
(AGI soon)
doomers
longtermists
AGI bad
Connor
imgflip.com


AI Alignment easy
AI Alignment hard
artificial stentience
Joscha Bach
scale is all you need
we should focus on the near term
we should focus on the long term

NOT SO FAST



STEPHEN HAWKING
Not afraid of black holes. A.I. is another story.



BILL GATES
First you'll lose your job. Then it gets scary.



STUART RUSSELL
Earth for the earthlings!



NICK BOSTROM
Prepare for "Disneyland without children."



MAX TEGMARK
Uh, can we talk about this?



DEMIS HASSABIS
Full speed ahead!



PETER THIEL
Will be a winner either way.



STEVE WOZNIAK
Resigned to being a robot's pet.



SAM ALTMAN
Sees intergalactic domination—or extinction.



ELON MUSK
Eyeing the next flight to Mars.



LARRY PAGE
Green-lighted Google Brain. 'Nuff said.



YANN LeCUN
Chill, people! We got this.



ANDREW NG
Trust the robot.



MARK ZUCKERBERG
Worried? Tell my A.I. butler.



HIT THE GAS

RAY KURZWEIL
Eager to be a cyborg.

Superintelligence

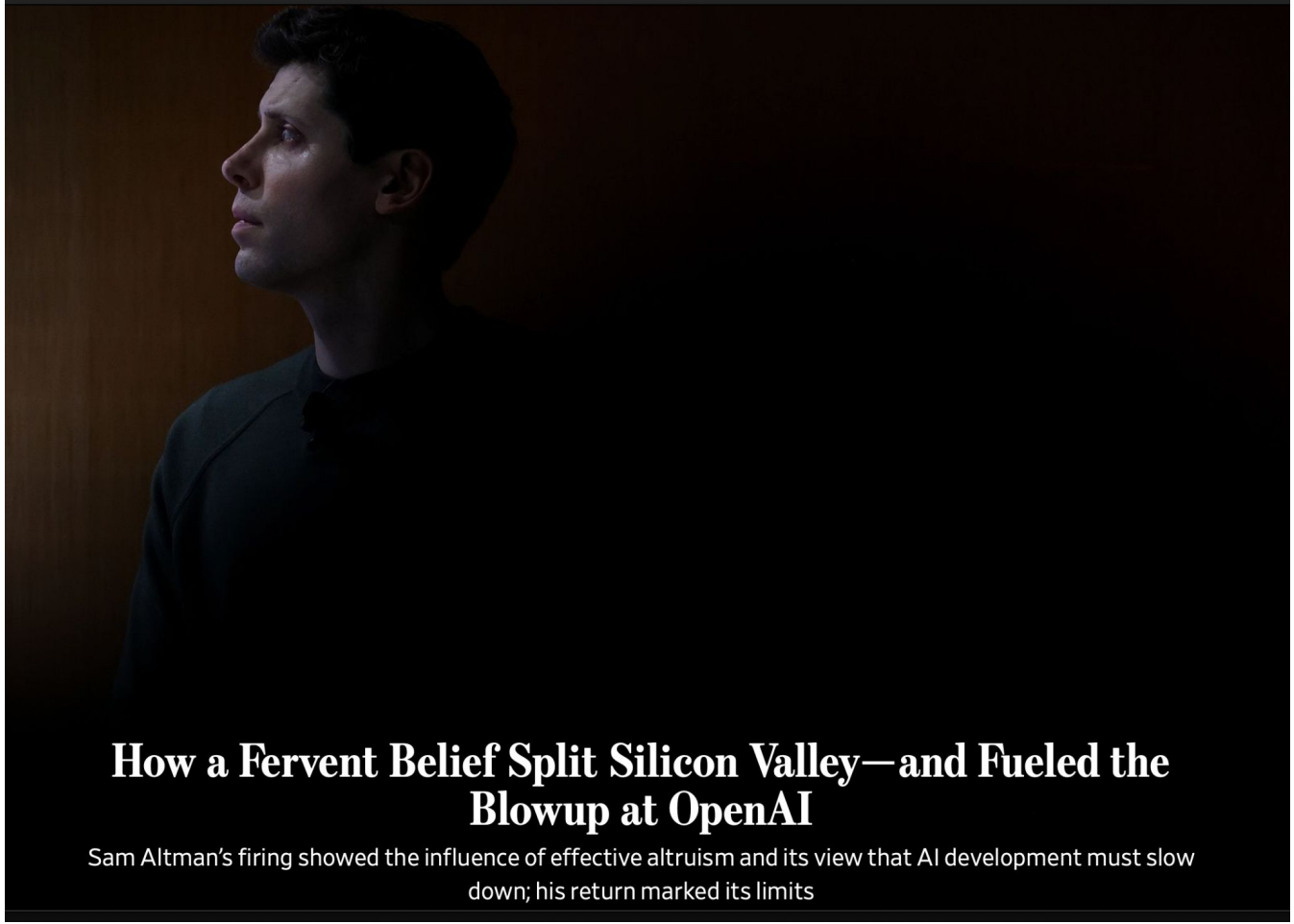
I. J. Good, mathematician 1960s:

> Let an ultraintelligent machine be defined as a machine that can far surpass all the intellectual activities of any any man however clever. Since the design of machines is one of these intellectual activities, an ultraintelligent machine could design even better machines; there would then unquestionably be an **"intelligence explosion,"** and the intelligence of man would be left far behind. Thus the first ultraintelligent machine is the _last_ invention that man need ever make, provided that the machine is docile enough to tell us how to keep it under control.
> ...
> It is more probable than not that, within the twentieth century, an ultraintelligent machine will be built and that it will be the last invention that man need make.



THE WALL STREET JOURNAL.

SIGN IN



How a Fervent Belief Split Silicon Valley—and Fueled the Blowup at OpenAI

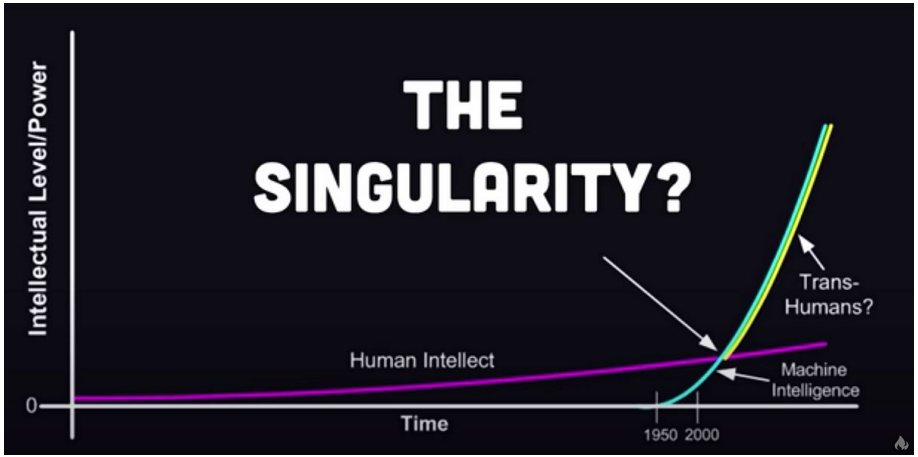
Sam Altman's firing showed the influence of effective altruism and its view that AI development must slow down; his return marked its limits

Singularity and Post-human Era, an old concept?

- In Math, “singularity” is when the function becomes undefined (e.g. a number is divided by zero).
- As such when AI becomes superhuman it surpasses human predictability, it reaches technological singularity
- Large computer networks “wake-up” to create superhuman entity
- Biological sciences will find ways to AI + humans

In the 1950s Stan Ulam paraphrased John von Neumann as saying:

One conversation centered on the ever accelerating progress of technology and changes in the mode of human life, which gives the appearance of approaching some **essential singularity** in the history of the race beyond which human affairs, as we know them, could not continue.



SQUAD 1.1 and SQUAD 2.0: F1 SCORE

Source: SQuAD 1.1 and SQuAD 2.0, 2021 | Chart: 2022 AI Index Report

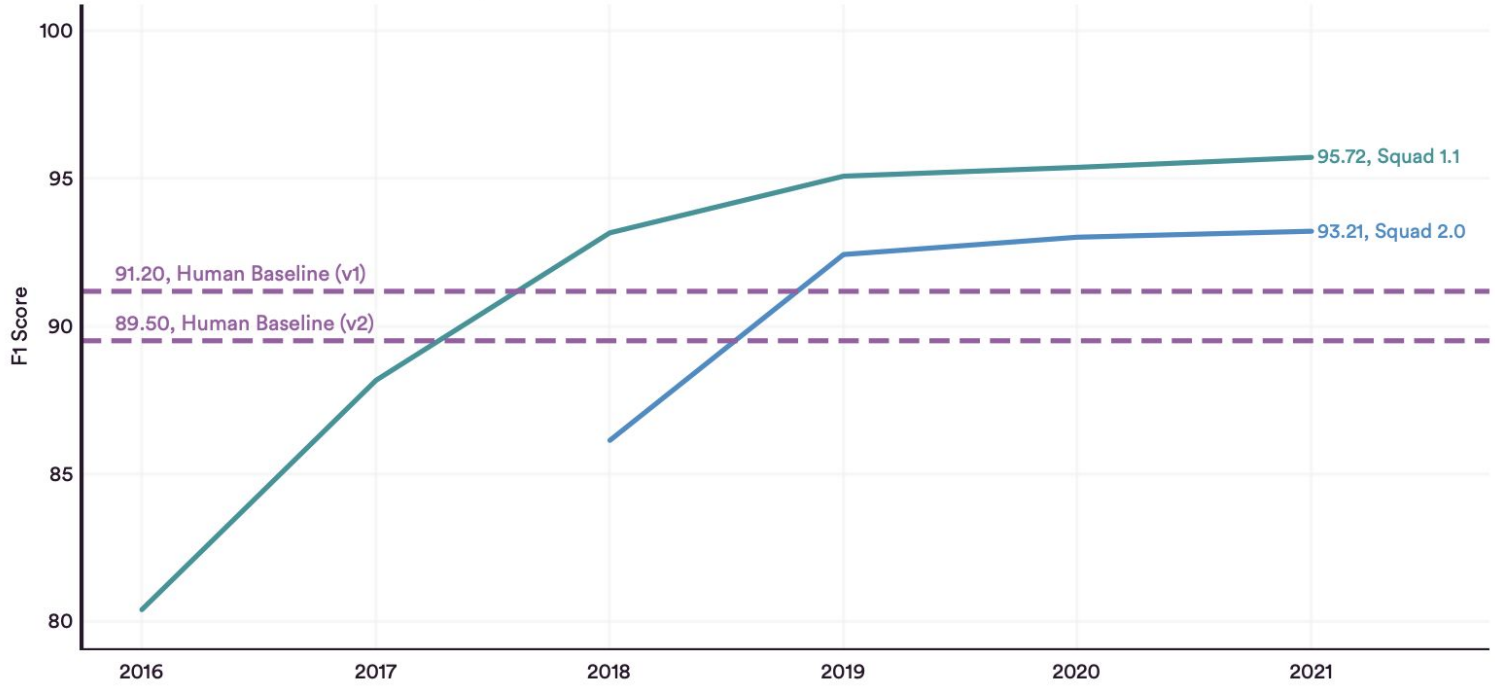


Figure 2.3.4

Singularity and Post-human Era 1990s

Vernor Vinge (Mathematician at San Diego State University) in 1993:

“Within thirty years, we will have the technological means to create superhuman intelligence. Shortly after, the human era will be ended.

Is such progress avoidable? If not to be avoided, can events be guided so that we may survive?”

“Even if all the governments of the world were to understand the ‘threat’ and be in deadly fear of it, progress toward the goal would continue. [...]. The competitive advantage – economic, military, even artistic – of every advance in automation is so compelling that passing laws, or having customs, that forbid such things merely assures that someone else will get them first.”

“If the Singularity can not be prevented or confined, just how bad could the **Post-Human era** be? **Well ... pretty bad. The physical extinction of the human race is one possibility.** Yet physical extinction may not be the scariest possibility. Again, analogies: **Think of the different ways we relate to animals. Some of the crude physical abuses are implausible, yet**....”

“In a Post-Human world there would still be plenty of niches where human equivalent automation would be desirable: embedded systems in autonomous devices, self-aware daemons in the lower functioning of larger sentients. (A strongly superhuman intelligence would likely be a Society of Mind [15] with some very competent components.) Some of these human equivalents might be used for nothing more than digital signal processing. They would be more like whales than humans. Others might be very human-like, yet with a one-sidedness, a dedication that would put them in a mental hospital in our era. Though none of these creatures might be flesh-and-blood humans, they might be the closest things in the new environment to what we call human now.”

“Accelerationist” vs. “Safetyist”

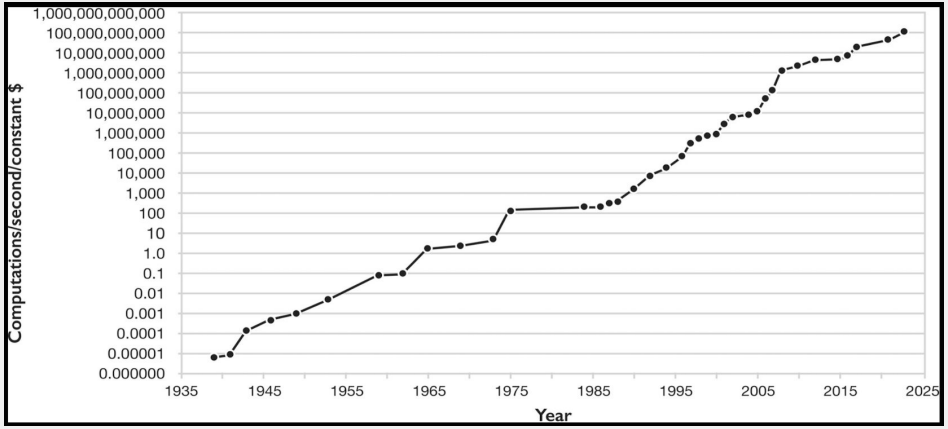
- Have to outcompete other powers (Russia, China)
- AI development speed-up lead to economic growth, scientific discovery, poverty reduction, higher education, etc.
- “move fast, break things”

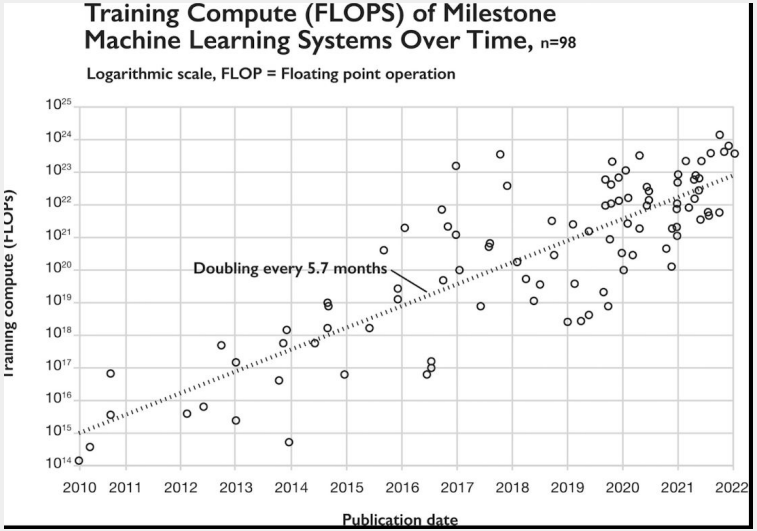
- Prioritize minimizing AI risk
- We could make mistakes that spiral out of control!
- “EA”-aligned
- Need to build robust safe-guards first

- AI/AGI could be greatest achievement ever or the biggest existential threat
- AI will self-improvement on a feedback loop, recursively

Ray Kurzweil

- Multiple engineering projects spanning text-to-speech, music synthesis, education, OCR
- Currently Director of Engineer (Google)
- Singularity is near (2005), Singularity is nearer (2024)?
- “Computers will pass Turing Test 2029”
- “We merge with AI and augment ourselves with millions of times the computational power that our biology gave us. This will expand our intelligence and consciousness so profoundly that it’s difficult to comprehend. This event is what I mean by the Singularity.”
- “rather than AI being a competitor, it will become an extension of ourselves.”
- “Tech advances are becoming cheaper and more nuanced at exponential rate”





3 Perils of Superintelligent AI (Kurzweil ver.)

- We can mitigate risk of superintelligent, runaway AI/AGI by being aware of the types of “errors”

3 Perils of Superintelligent AI (Kurzweil ver.)

- We can mitigate risk of superintelligent, runaway AI/AGI by being aware of the types of “errors”

1) Misuse: AI is implemented correctly and functions as human operators had intended but the human operators use it for malicious means.

    e.g. terrorists might use an AI’s biochemistry abilities to design a new virus that causes a deadly pandemic.

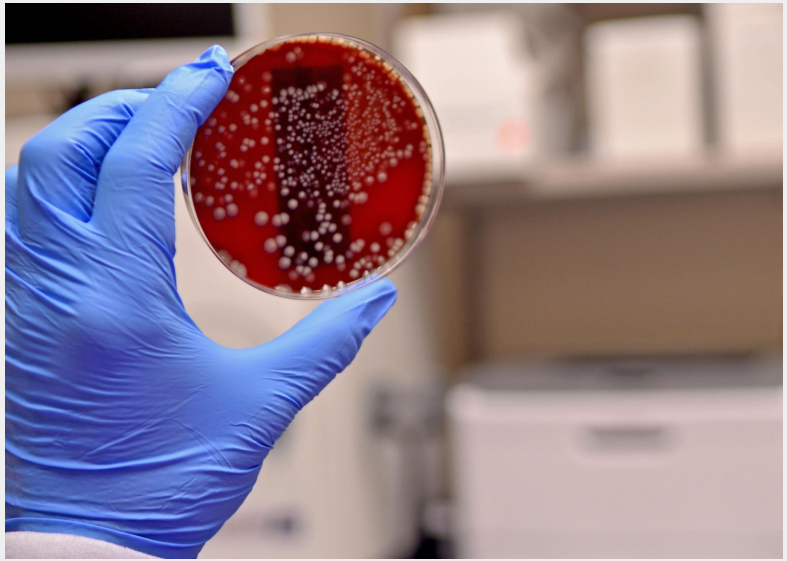


3 Perils of Superintelligent AI (Kurzweil ver.)

- We can mitigate risk of superintelligent, runaway AI/AGI by being aware of the types of “errors”

2) Outer-Misalignment: mismatch between the programmers’ actual intentions and the goals they teach the AI in hopes of achieving them.

e.g. AI is taught to kill all cancer cells based on a DNA mutation signal but that signal is present in many healthy cells too.

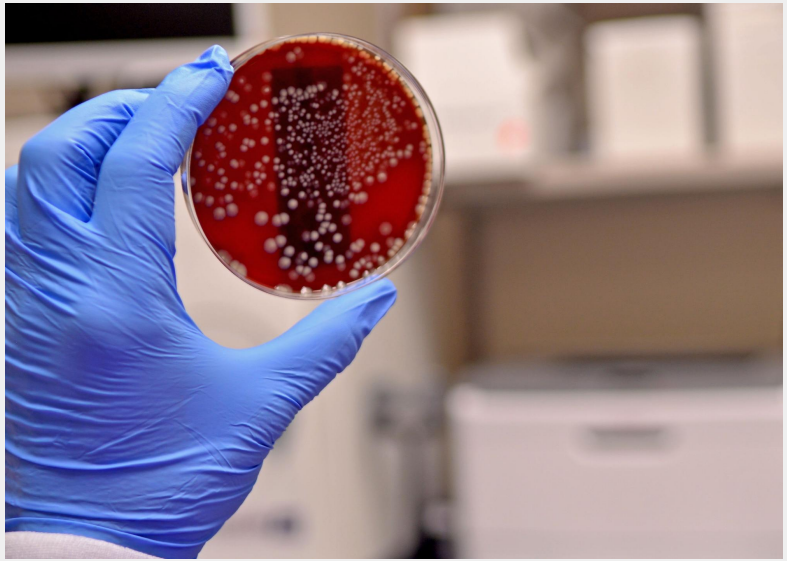


3 Perils of Superintelligent AI (Kurzweil ver.)

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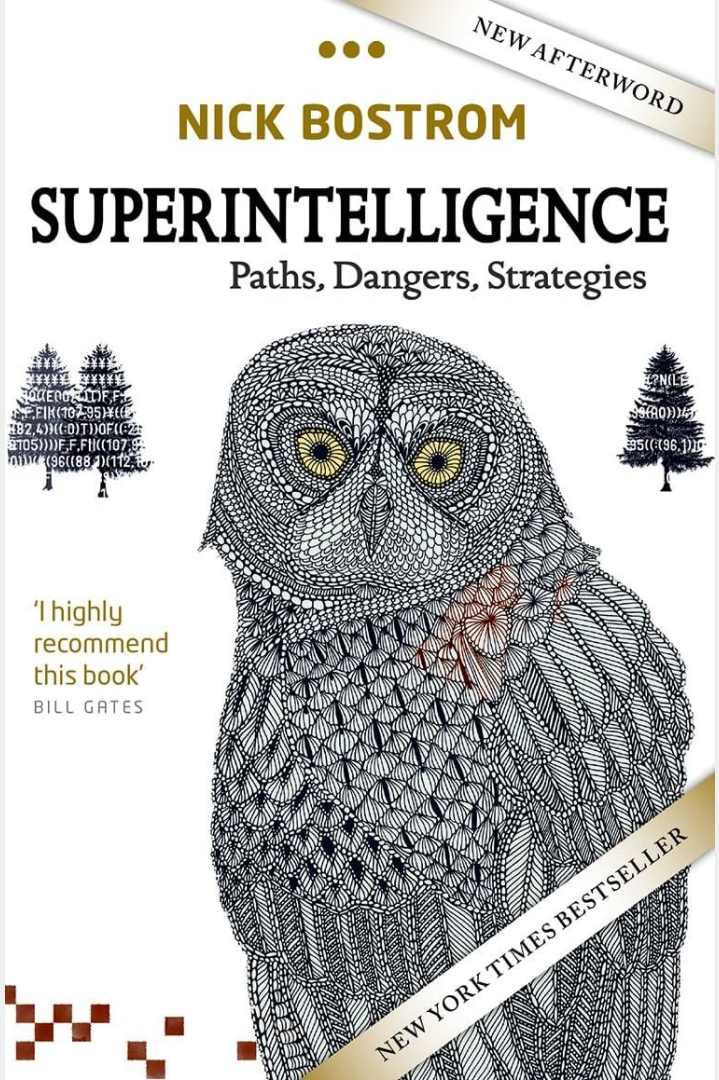
2) Inner-Misalignment: the methods the AI learns to achieve its goal produce undesirable behavior, at least in some cases.

e.g. AI is highly effective on sample cancer cells but not for all cancer cells – ends up killing healthy cells.



Nick Bostrom

- Founding Director at Future of Humanity Institute (2005-2024)
- Superintelligence: Paths, Dangers, Strategies (2014)
- “In the 21st century, we could develop recursively self-improving AI that could be out of control”
- “Have to value-align AI so that it does what humans want it to”
- Introduces AI safety frameworks such as “boxing” (containment) and value-loading



Bostrom’s Paperclip Experiment!

- Bostrom’s 2003 paper "Ethical Issues in Advanced Artificial Intelligence."
- Introduces the concept of AI Alignment (misalignment)



Maximize Paperclip Production!

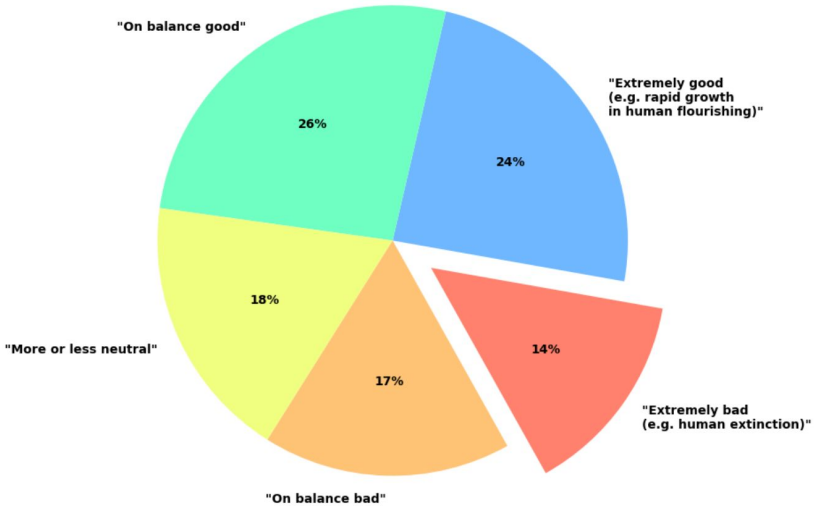


2022 Expert Survey on Progress in AI

- 2022 ESPAI is a survey of machine learning researchers that AI Impacts ran in June-August 2022
- 738 responses, some partial, for a 17% response rate from, researchers who published at the conferences NeurIPS or ICML in 2021
- ‘HLMI’ was defined as follows:
  The following questions ask about ‘high–level machine intelligence’ (HLMI). Say we have ‘high-level machine intelligence’ when ***unaided machines can accomplish every task better and more cheaply than human workers****. Ignore aspects of tasks for which being a human is intrinsically advantageous, e.g. being accepted as a jury member. Think feasibility, not adoption.*

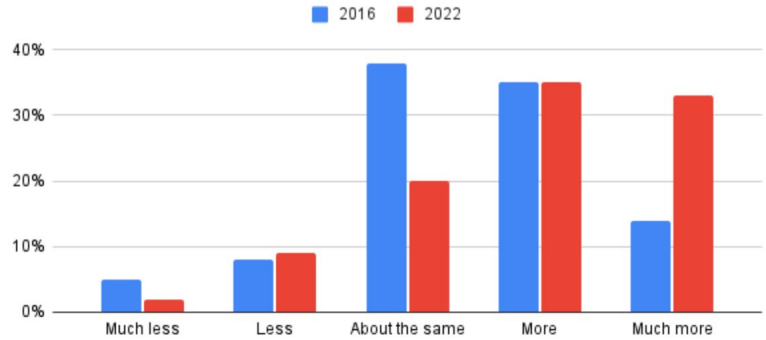
**"Assume for the purpose of this question that HLMI will at some point exist.
How positive or negative do you expect the overall impact of this to be on humanity, in the long run?"**

Average responses from 559 machine learning experts in 2022



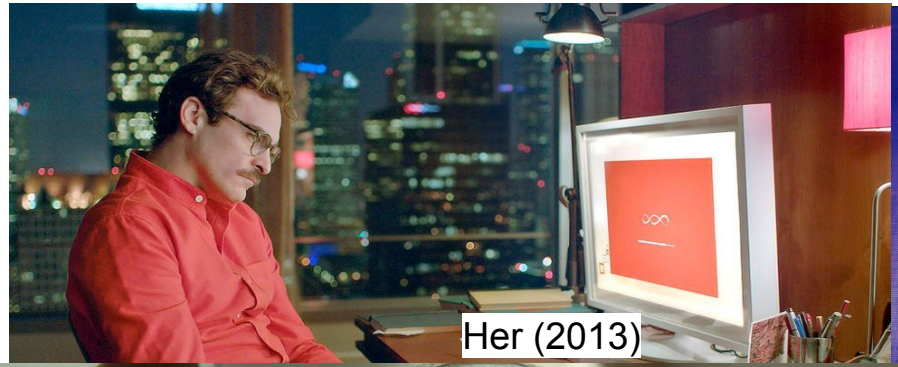
"How much should society prioritize AI safety research, relative to how much it is currently prioritized?"

Responses from machine learning experts in 2016 and 2022

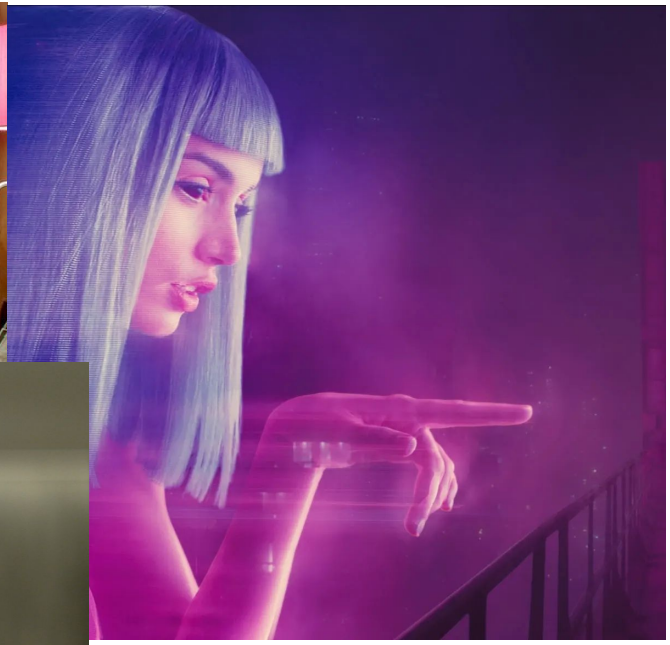


- **The aggregate forecast time to a 50% chance of HLMI was 37 years, i.e. 2059** (not including data from questions about the conceptually similar Full Automation of Labor, which in 2016 received much later estimates)**.** This timeline has become about eight years shorter in the six years since 2016, when the aggregate prediction put 50% probability at 2061, i.e. 45 years out. Note that these estimates are conditional on “human scientific activity continu[ing] without major negative disruption.”
- **The median respondent believes the probability that the long-run effect of advanced AI on humanity will be “extremely bad (e.g., human extinction)” is 5%.** This is the same as it was in 2016 (though Zhang et al 2022 found 2% in a similar but non-identical question). Many respondents were substantially more concerned: 48% of respondents gave at least 10% chance of an extremely bad outcome. But some much less concerned: 25% put it at 0%.
- **The median respondent believes society should prioritize AI safety research “more” than it is currently prioritized.** Respondents chose from “much less,” “less,” “about the same,” “more,” and “much more.” 69% of respondents chose “more” or “much more,” up from 49% in 2016.
- **The median respondent thinks there is an “about even chance” that a stated argument for an intelligence explosion is broadly correct.** 54% of respondents say the likelihood that it is correct is “about even,” “likely,” or “very likely” (corresponding to probability >40%), similar to 51% of respondents in 2016. The median respondent also believes machine intelligence will probably (60%) be “vastly better than humans at all professions” within 30 years of HLMI, and the rate of global technological improvement will probably (80%) dramatically increase (e.g., by a factor of ten) as a result of machine intelligence within 30 years of HLMI.

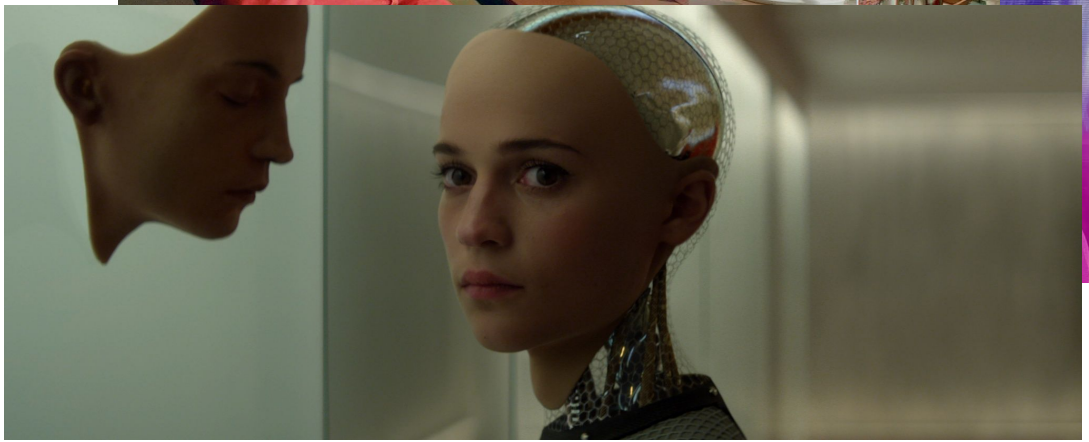
On the other hand, 21st C romance with AI?



Her (2013)



Blade Runner (2017)



Ex Machina (2014)

FROM THE MAGAZINE APRIL 2017 ISSUE

Elon Musk's Billion-Dollar Crusade to Stop the A.I. Apocalypse

Elon Musk is famous for his futuristic gambles, but Silicon Valley's latest rush to embrace artificial intelligence scares him. And he thinks you should be frightened too. Inside his efforts to influence the rapidly advancing field and its proponents, and to save humanity from machine-learning overlords.

BY MAUREEN DOWD

MARCH 26, 2017

"WE ARE SUMMONING THE DEMON."



Halt AI development!

Pause Giant AI Experiments: An Open Letter

We call on all AI labs to immediately pause for at least 6 months the training of AI systems more powerful than GPT-4.

Signatures
33709

Add your signature

Published
March 22, 2023

AI systems with human-competitive intelligence can pose profound risks to society and humanity, as shown by extensive research[1] and acknowledged by top AI labs.[2] As stated in the widely-endorsed Asilomar AI Principles, *Advanced AI could represent a profound change in the history of life on Earth, and should be planned for and managed with commensurate care and resources*. Unfortunately, this level of planning and management is not happening, even though recent months have seen AI labs locked in an out-of-control race to develop and deploy ever more powerful digital minds that no one – not even their creators – can understand, predict, or reliably control.

Contemporary AI systems are now becoming human-competitive at general tasks,[3] and we must ask ourselves: *Should* we let machines flood our information channels with propaganda and untruth? *Should* we automate away all the jobs, including the fulfilling ones? *Should* we develop

Halt AI development!

Therefore, **we call on all AI labs to immediately pause for at least 6 months the training of AI systems more powerful than GPT-4**. This pause should be public and verifiable, and include all key actors. If such a pause cannot be enacted quickly, governments should step in and institute a moratorium.

"The time for saying that this is just pure research has long since passed. [...] It's in no country's interest for any country to develop and release AI systems we cannot control. Insisting on sensible precautions is not anti-industry. Chernobyl destroyed lives, but it also decimated the global nuclear industry. I'm an AI researcher. I do not want my field of research destroyed. Humanity has much to gain from AI, but also everything to lose."

Stuart Russell, *Smith-Zadeh Chair in Engineering and Professor of Computer Science at the University of California, Berkeley, Founder of the Center for Human-Compatible Artificial Intelligence (CHAI).*

"Let's slow down. Let's make sure that we develop better guardrails, let's make sure that we discuss these questions internationally just like we've done for nuclear power and nuclear weapons. Let's make sure we better understand these very large systems, that we improve on their robustness and the process by which we can audit them and verify that they are safe for the public."

Yoshua Bengio, *Scientific Director of the Montreal Institute for Learning Algorithms (MILA), Professor of Computer Science and Operations Research at the Université de Montréal, 2018 ACM A.M. Turing Award Winner.*

"We have a perfect storm of corporate irresponsibility, widespread adoption, lack of regulation and a huge number of unknowns. [FLI's Letter] shows how many people are deeply worried about what is going on. I think it is a really important moment in the history of AI - and maybe humanity."

Gary Marcus, *Professor Emeritus of Psychology and Neural Science at New York University, Founder of Geometric Intelligence*

"It feels like we are moving too quickly. I think it is worth getting a little bit of experience with how they can be used and misused before racing to build the next one. This shouldn't be a race to build the next model and get it out before others."

Peter Stone, *Professor at the University of Texas at Austin, Chair of the One Hundred Year Study on AI.*

"We don't know what these [AI] systems are trained on or how they are being built. All of this happens behind closed doors at commercial companies. This is worrying."

Catelijne Muller, *President of ALLAI, Member of the EU High Level Expert Group on AI*

"Those making these [AI systems] have themselves said they could be an existential threat to society and even humanity, with no plan to totally mitigate these risks. It is time to put commercial priorities to the side and take a pause for the good of everyone to assess rather than race to an uncertain future"

Emad Mostaque, *Founder and CEO of Stability AI*



Policy recommendations:

1. Mandate robust third-party auditing and certification.
2. Regulate access to computational power.
3. Establish capable AI agencies at the national level.
4. Establish liability for AI-caused harms.
5. Introduce measures to prevent and track AI model leaks.
6. Expand technical AI safety research funding.
7. Develop standards for identifying and managing AI-generated content and recommendations.

“We can see only a short distance ahead, but we can see that much remains to be done.”

- Alan Turing (1950)

